

MTH1001: MATHEMATICAL STRUCTURES

Summary of the module

Term 1, 2025–2026

This module is an introduction to university-level Pure Mathematics.

Pure mathematics is concerned with developing the theory of abstract mathematical objects (numbers, functions, sets, etc) by starting from precise definitions and proving general results about them. In order to understand and construct mathematical proofs, we need to think carefully about how to write down logically correct arguments. We will therefore emphasise *reading* and *writing* mathematics, as well as learning methods to answer specific types of questions. Some of the topics we will cover also appear in other modules, but from a different point of view.

The topics we will cover in Term 1 are

- sets and logic

- functions

- number theory (prime factorisations, etc)

- groups

The syllabus plan/reading list can be found in the [module descriptor](#):

Staff details

Nigel Byott (term 1)

N.P.Byott@exeter.ac.uk

Harrison 320

Office hours: Mondays 15.30-16.30, Harrison 320

Up to Reading Week: Fridays 13.30-14.30,

Harrison 206 (LRC): Combined Stage 1 Office Hour

After Reading Week: Fridays 10.30-11.30, Harrison 320

Pete Ashwin (term 2, weeks 1-5)

P.Ashwin@exeter.ac.uk

Harrison 338

Office hours: Thursdays 10.30-11.30,

Harrison 206 (LRC): Combined Stage 1 Office Hour

Jan Sieber (term 2, weeks 6-11)

J.Sieber@exeter.ac.uk

Harrison 333

Office hours: Thursdays 10.30-11.30,

Harrison 206 (LRC): Combined Stage 1 Office Hour

Department of Mathematics & Statistics, University of Exeter

November 26, 2025

This document is typeset using the `tufte-latex` package. Macros from the Legrand Orange template at <http://www.LaTeXTemplates.com> have been incorporated into the `tufte-handout-local.tex` file.

Contents

1	SETS AND LOGIC	6
1.1	<i>Sets</i>	6
1.1.1	<i>Set constructor notation:</i>	7
1.1.2	<i>The empty set</i>	8
1.1.3	<i>Notation for standard systems of numbers</i>	8
1.1.4	<i>Subsets</i>	9
1.1.5	<i>Operations on sets</i>	9
1.1.6	<i>The powerset of a set</i>	11
1.1.7	<i>Cartesian products</i>	12
1.2	<i>Writing proofs with sets</i>	12
1.2.1	<i>Proving one set is a subset of another</i>	13
1.2.2	<i>Proving two sets are equal</i>	15
1.3	<i>Propositional Calculus</i>	18
1.3.1	<i>Propositions</i>	18
1.3.2	<i>Quantifiers</i>	20
1.3.3	<i>Operations on Propositions</i>	21
1.3.4	<i>Identities for compound propositions</i>	22
1.3.5	<i>De Morgan's Laws</i>	24
1.3.6	<i>Negating Quantified Statements</i>	26
1.3.7	<i>Implication</i>	27
1.3.8	<i>Deduction</i>	29
1.4	<i>General Methods of Proof</i>	31
1.4.1	<i>Direct Proof</i>	31
1.4.2	<i>Contrapositive Proof</i>	32
1.4.3	<i>Proof by Contradiction</i>	32

2	<i>FUNCTIONS</i>	34
2.1	<i>What is a function?</i>	34
2.2	<i>Surjective and Injective Functions</i>	38
2.3	<i>The “algebra” of functions</i>	41
2.3.1	<i>Composition of functions</i>	41
2.3.2	<i>Identity Functions</i>	44
2.3.3	<i>Inverse functions</i>	44
2.4	<i>Permutations</i>	50
2.4.1	<i>Cycle notation</i>	53
2.4.2	<i>Composing permutations in cycle notation</i>	55
2.4.3	<i>The inverse of a permutation</i>	56
2.4.4	<i>The order of a permutation</i>	57
2.4.5	<i>The sign of a permutation</i>	59
3	<i>ELEMENTARY NUMBER THEORY</i>	62
3.1	<i>Proof by Induction</i>	62
3.2	<i>Prime Numbers and Divisibility</i>	70
3.3	<i>Modular arithmetic</i>	77
3.4	<i>Greatest Common Divisors</i>	84
3.5	<i>Uniqueness of Prime Factorisation</i>	91
4	<i>GROUPS</i>	95
4.1	<i>Binary operations</i>	95
4.2	<i>Axioms of a group</i>	96
4.3	<i>Dihedral groups</i>	101
4.4	<i>The cardinality of a group and the order of an element</i>	105

4.5	<i>Subgroups</i>	107
4.6	<i>Cosets and Lagrange's Theorem</i>	111
4.7	<i>Isomorphisms and homomorphisms</i>	114

1 SETS AND LOGIC

1.1 Sets

Virtually all of mathematics can be formulated in terms of sets, so we take the idea of a set as our starting point. In fact, the notion of a set is so important that you will see it in some of your other modules as well.

A **set** is (almost¹) any collection of objects. The objects in a set are called its **elements** or **members**.

¹ A set cannot be an elements of itself, so there is not such thing as the set of all sets.

We can specify a set by listing its elements inside braces (curly brackets) $\{\dots\}$, so if

$$A = \{\text{John}, \text{Susan}, \text{Ahmed}\}$$

and

$$B = \{\text{Exeter}, \text{Newcastle}, \text{Sheffield}, \text{York}\}$$

then A is a set of people and B is a set of English cities.

Repeats in the list of elements in a set, or changes in the order, do not affect the set. All that matters is which objects are elements and which are not, rather than the way we list them. So we could also the above sets as

$$A = \{\text{Susan}, \text{John}, \text{Susan}, \text{Ahmed}, \text{Ahmed}\}$$

and

$$B = \{\text{York}, \text{Newcastle}, \text{Exeter}, \text{Sheffield}\}.$$

We write $|S|$ for the number of elements in a set S (which might be infinite), and call this the **cardinality** (or **size**) of S . So $|A| = 3$ and $|B| = 4$.

We write $x \in A$ to mean that x is an element of the set A , and $x \notin$

A to mean x is not an element of A , so $Susan \in A$ but $Manchester \notin B$.

The elements in a set can be any sort of objects, but we will usually be considering sets of mathematical objects (numbers, points, vectors, functions, or even other sets). For example,

$$\{2, \pi, \sqrt{7}, e^\pi\}$$

is a set of real numbers. On the other hand,

$$\{\{1, 2, 3\}, \{4, 5, 6, 7\}\}$$

is a set of sets. It has two elements, namely the set $\{1, 2, 3\}$ and the set $\{4, 5, 6, 7\}$. It is not the same thing as the set

$$\{1, 2, 3, 4, 5, 6, 7\}$$

(which is a set of numbers with 7 elements). Usually the elements in a set are all the same sort of object, but we can also have sets like

$$S = \{11, \{6, 7, 8, 9\}, Susan\}.$$

This has three elements: the number 11, the set $\{6, 7, 8, 9\}$ and the person Susan. So the number 6 is *not* an element of the set S ; it is an element of an element of S . As these examples show, it is important to look carefully at where the braces occur.

1.1.1 Set constructor notation:

Another way to specify a set is to obtain its elements in some way from an existing set. For example, if

$$A = \{3, 5, 6, 8, 9\}$$

we can define the sets

$$B = \{n \in A : n \text{ is odd}\},$$

$$C = \{\sqrt{n} : n \in A\}.$$

Then

$$B = \{3, 5, 9\}, \quad C = \{\sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{8}, 3\}.$$

Notice that it is the objects *before* the colon which are the elements of the new set.

1.1.2 The empty set

This is the set with no elements at all. We could write it $\{\}$, but we usually use the special symbol $\emptyset$.

1.1.3 Notation for standard systems of numbers

$$\mathbb{N} = \{1, 2, 3, \dots\} \text{ (the natural number).}$$

(Some people take $\mathbb{N}$ to start at 0.)

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\} \text{ (the integers);}$$

$$\mathbb{Q} = \{n/m : n, m \in \mathbb{Z} \text{ with } m \neq 0\} \text{ (the rational numbers) ;}$$

$$\mathbb{R} \text{ (the real numbers) .}$$

We can think of real numbers (somewhat imprecisely) as infinite decimals, or as points on the "number line". There are many real numbers, such as π and $\sqrt{2}$, which are not rational. Don't confuse the last two: the letter $\mathbb{R}$ stands for *real*, not *rational*. The letter $\mathbb{Q}$ is for *quotient* (another word for *fraction*).

Another standard number system is the complex numbers.

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$$

where i is defined so that $i^2 = -1$. If you haven't come across the complex numbers before, you will learn about them in MTH1002 towards the end of term 1. We won't use complex numbers during term 1 in MTH1001.

1.1.4 Subsets

Definition 1.1 — Subset. Let A be a set. A *subset* of A is a set B such that every element of B is an element of A .

We write $B \subseteq A$ (or $B \subset A$) to mean that B is a subset of A .

For any set A , we have $A \subseteq A$ (every element of A is an element of A) and $\emptyset \subseteq A$ (we can't find an element of $\emptyset$ which is not in A , since we can't find an element of A at all).

The set $A = \{1, 2, 3\}$ has 8 subsets: These are

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}.$$

In general, if a set A has n elements, then it has 2^n subsets because, in order to choose a subset B of A we must choose to include or exclude each of the n elements in turn.

Recall that all that matters about a set is what elements it has. So if A and B are sets with exactly the same elements, then they are the same set. Let us state this a bit more formally, using Definition 1.1.

Definition 1.2 — Equality of sets. Let A and B be sets. Then they are *equal* (written $A = B$) if $A \subseteq B$ and $B \subseteq A$.

This just says that $A = B$ precisely when every element of A is an element of B , and also every element of B is an element of A .

1.1.5 Operations on sets

We can combine sets A and B in various ways:

Definition 1.3 Let A and B be sets. Then

(i) The **union** of A and B is

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

the set of elements which are in at least one of A and B . (Here, as usually in mathematics, “or” is to be interpreted in the inclusive sense: $x \in A$ or $x \in B$ or *both*.)

(ii) The **intersection** of A and B is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

the set of elements which are in both A and B .

(iii) The (set-theoretic difference) of A and B is

$$A \setminus B = \{x \in A : x \notin B\},$$

the set of elements which are in A but are not in B .

Similarly, we can form the union and intersection of more than 2 sets.

■ **Example 1.4** Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{4, 5, 6\}$. Then

$$A \cap B = \{2, 3\}, \quad B \cup C = \{2, 3, 4, 5, 6\}, \quad C \setminus B = \{5, 6\},$$

$$A \cup B \cup C = \{1, 2, 3, 4, 5, 6\}, \quad A \cap B \cap C = \emptyset, \quad A \cup (B \cap C) = \{1, 2, 3, 4\}.$$

Here is a more complicated example.

■ **Example 1.5** Let $S = \{3, 4, 5\}$ and $T = \{1, 2, \{3, 4\}, 5\}$. Then

$$S \cap T = \{5\}, \quad \{3, 4\} \subseteq S \text{ but } \{3, 4\} \notin S.$$

On the other hand,

$$\{3,4\} \in T, \text{ but } \{3,4\} \notin T.$$

Remark 1.6

You might be used to thinking about unions and intersections of sets pictorially using Venn diagrams. This is often a helpful way to understand statements about sets, but when we want to write down proofs we need to use the definitions of these operations as given above. A Venn diagram does not constitute a proof. In this module, I will deliberately avoid drawing Venn diagrams.

Remark 1.7

If all the sets we are considering are subsets of a fixed set S (say the real numbers), then we refer to this set S as the *universal set*, and for a subset A of S , we define the complement A^c of A in S to be the set of elements of S not in A , so

$$A^c = \{s \in S : s \notin A\} = S \setminus A.$$

The notion of a complement only makes sense once we have fixed an appropriate universal set.

1.1.6 The powerset of a set

The **powerset** of a set S , denoted by $\mathcal{P}(S)$, is the set of all subsets of S . (It is a set of sets.) So for $A = \{1, 2, 3\}$ we have

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

In general, if S is a finite set with $|S| = n$ then $\mathcal{P}(S) = 2^n$ since S has 2^n subsets.

1.1.7 Cartesian products

If A and B are sets, we write $A \times B$ for the set of **ordered pairs**² (a, b) with $a \in A$ and $b \in B$. (These are *ordered* pairs because we treat (b, a) as different from (a, b) .) The set $A \times B$ is called the **Cartesian product** (or just the **product**) of A and B . If $B = A$ we often write A^2 instead of $A \times A$. For example if $A = \{3, 4, 5\}$ and $B = \{\text{London}, \text{Manchester}\}$, then

$$A \times B = \{(3, \text{London}), (4, \text{London}), (5, \text{London}), \\ (3, \text{Manchester}), (4, \text{Manchester}), (5, \text{Manchester})\},$$

a set with 6 elements, and

$$A^2 = \{(3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)\},$$

a set with 9 elements. In general, if A and B are finite sets with $|A| = m$ and $|B| = n$ then $|A \times B| = mn$ since in order to choose an element (a, b) of $A \times B$ we have m choices for a and n choices for b .

² Don't confuse this notation for ordered pairs with the notation for open intervals in the real numbers, such $(c, d) = \{x \in \mathbb{R} : c < x < d\}$. Brackets are used for many different things in mathematics, but it is usually obvious from the context which meaning is intended.

1.2 Writing proofs with sets

In mathematics, we establish that new results are true by giving proofs of them. A proof is a logically correct argument explaining why the claimed result must necessarily be true. Since it is an *argument* rather than just a *calculation*, it needs to be set out carefully, using mathematical terminology clearly and accurately in order to explain the reasoning.

In this section, we will think about how we actually *write down* proofs. The statements we will prove are fairly simple (you might even consider them obvious), since the focus for the moment is on what we actually need to write in order to prove them.

1.2.1 Proving one set is a subset of another

Suppose we are given sets A and B and asked to *prove* that $A \subseteq B$. Where do we begin? The first thing is ask what it means for A to be a subset of B . We know the answer to that by Definition 1.1: it means that every element of A is an element of B . The general pattern of the proof is forced upon us by the definition: we need to start with an arbitrary element of A , and explain why it is an element of B . By *arbitrary* I mean that our explanation has to be valid whichever element of A we started with. Here is an example.

■ **Example 1.8** Let $A = \{x \in \mathbb{R} : x^2 < 5\}$ and let $B = \{x \in \mathbb{R} : x \leq 4\}$. Prove that $A \subseteq B$.

Proof. Let $x \in A$.

Then $x^2 < 5$.

Therefore $-\sqrt{5} < x < \sqrt{5}$.

But $\sqrt{5} \leq 4$.

Thus $x \leq 4$.

Hence $x \in B$.

So every element of A is an element of B , showing that $A \subseteq B$. ■

Let's notice several things about this proof.

- We have indicated the logical connections between the various statements in the proof by linking them with words such as *Let*, *If ... then ...*, *therefore* (or corresponding symbols like $\therefore$). This means writing in sentences: a sequence of equations, with no words in between, does not *explain* anything.
- Within the proof, we have made use of standard properties of real numbers. In particular, we used the transitivity of the order relation $<$: if $x < y$ and $y < z$ then $x < z$ (with $y = \sqrt{5}$ and $z = 4$). We will look in more detail at the real numbers in Term 2, but

for the moment I will assume you are familiar with manipulating inequalities.

- You might wonder why I wrote $\sqrt{5} \leq 4$ rather than $\sqrt{5} < 4$. Both statements are true (for any real numbers u, v , if $u < v$ then it is certainly true that $u \leq v$) but since the condition for $x \in B$ involves $\leq$, we are going to need $\leq$ in our proof.

The proof does not have to be written in *exactly* the way shown above, but it does have to explain the reasoning clearly and correctly. It doesn't really matter if we write "Let $x \in A$ " or "Suppose that x is an element of A ", or if we write $\therefore$ instead of "therefore". We don't have to use x to label our element of A : any other letter would do (as long as it is not already in use for something else). The following would be a perfectly acceptable proof.

Proof.

Suppose that y is an element of A , so $y^2 < 5$.

$$\therefore y < \sqrt{5}.$$

Since $\sqrt{5} < 4$, we have $y < 4$, so $y \leq 4$.

This means $y \in B$.

Hence every element of A is an element of B , so $A \subseteq B$. ■

On the other hand, the following is not an acceptable proof, and would not get many marks in a test or exam, even if it is the sort of thing you might scribble down as rough work.

Proof.

[Don't do this!]

$$y^2 < 5$$

$$y < \sqrt{5}$$

$$\rightarrow y < 4.$$



What is wrong with it?

- The symbol y needs to be introduced, e.g. with "Let $y \in A$ " or "If y is a real number then ...". Otherwise, when I read the proof I will not know what it represents: an element of A ? any real number? an integer?
- There is nothing to say how the second line is related to the first. If you are telling me that the second line *follows from* the first, you need to indicate this, e.g. by "Hence" or $\therefore$.
- The last line misuses the symbol $\rightarrow$. Don't confuse $\rightarrow$ (which means "tends to", as in $1/n \rightarrow 0$ as $n \rightarrow \infty$) with $\Rightarrow$ (which means "implies"). Writing the last line as $\Rightarrow y < 4$ would be better, but still does not quite express the reasoning correctly: the point is not that *if* $y < \sqrt{5}$ *then* $y < 4$, but that *we already know that* $y < \sqrt{5}$ *so it is true that* $y < 4$.

1.2.2 Proving two sets are equal

Definition 1.2 tells us what it means for sets A and B to be equal: $A = B$ means precisely that $A \subseteq B$ and $B \subseteq A$. So we want to prove that $A = B$, we have two tasks: prove that $A \subseteq B$ and prove that $B \subseteq A$. Each of these parts will follow the pattern we have just seen.

For the next example, we will use the following definitions:

Definition 1.9 Let $n \in \mathbb{Z}$. Then n is *even* if $n = 2m$ for some $m \in \mathbb{Z}$, and n is *odd* if $n = 2m + 1$ for some $m \in \mathbb{Z}$.

Every integer is either even or odd.

■ **Example 1.10** Let

$$A = \{n \in \mathbb{Z} : n \text{ is even}\}$$

and

$$B = \{u + v : u, v \in \mathbb{Z} \text{ and } u, v \text{ are both odd}\}.$$

Prove that $A = B$.

Here, $n \in B$ precisely when n can be written (in at least one way) as $n = u + v$ for odd integers u, v . So, for example, $8 \in B$ since $8 = 3 + 5$. There are many other way to write 8 in this form, such as $8 = (-7) + 15$. Indeed, there are infinitely many ways to write any $n \in B$ in the required form.

Proof.

To prove that $A \subseteq B$:

Let $n \in A$.

Then $n = 2m$ for some $m \in \mathbb{Z}$.

Let $u = 1 = 2 \times 0 + 1$ and let $v = 2m - 1 = 2(m - 1) + 1$.

Then u and v are odd, and $n = u + v$, so $n \in B$.

Hence $A \subseteq B$.

To prove that $B \subseteq A$:

Let $n \in B$.

Then we can write $n = u + v$ for some odd integers u, v .

Since u is odd, we have $u = 2k + 1$ for some $k \in \mathbb{Z}$.

Since v is odd, we have $v = 2h + 1$ for some $h \in \mathbb{Z}$.

Then $n = u + v = 2(k + h + 1)$.

Since $k + h + 1 \in \mathbb{Z}$, this shows that n is even.

Hence $B \subseteq A$.

Since $A \subseteq B$ and $B \subseteq A$, we have $A = B$.

■

Let's do a more general example of proving equality between sets.

■ **Example 1.11** Let A, B, C be any subsets of a set S . Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

We might convince ourselves this is true by drawing Venn diagrams, but that does not count as a proof. To write down a proof, need to recall the definitions

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Proof. Let $x \in A \cup (B \cap C)$. Then either $x \in A$ or $x \in B \cap C$.

If $x \in B \cap C$, then $x \in B$.

So either $x \in A$ or $x \in B$. Thus $x \in A \cup B$.

Similarly, since $x \in A$ or $x \in B \cap C$, we have $x \in A \cup C$.

So $x \in A \cup B$ and $x \in A \cup C$, which means $x \in (A \cup B) \cap (A \cup C)$.

Hence $x \in (A \cup B) \cap (A \cup C)$.

This shows that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Now let $x \in (A \cup B) \cap (A \cup C)$.

Then $x \in A \cup B$ and $x \in A \cup C$.

So either $x \in A$ or $x \in B$, and also either $x \in A$ or $x \in C$.

Thus if $x \notin A$ then $x \in B$ and $x \in C$ (that is, $x \in B \cap C$).

So either $x \in A$, or $x \in B \cap C$.

Hence $x \in A \cup (B \cap C)$.

This shows that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Since each of the sets $A \cup (B \cap C)$ and $(A \cup B) \cap (A \cup C)$ is a subset of the other, they are equal. ■

In this proof, we have had to reason with simple statements like $x \in A$ and $x \in B$ connected with *and* and *or*. The proof relies on being able to rewrite these compound statements in an equivalent

form. We might describe this sort of reasoning as “common sense”, but it is easy to get muddled when doing this. In the next subsection, we will think about how to work with these sorts of statements more systematically.

1.3 Propositional Calculus

Now that we have seen some simple examples of proofs, we will look more closely at logical reasoning in mathematics.

1.3.1 Propositions

In mathematics, a *proposition* is a statement which is either true or false (but not both!) It is not allowed to be anything in between, such as *more-or-less true* or *a matter of opinion*. Consider the following examples:

- (i) $4 + 3 = 7$.
- (ii) All cats are grey.
- (iii) $x^2 > 2$ for every real number x .
- (iv) Sit down!
- (v) Spinach is delicious.
- (vi) $\sqrt{2} = 1.414$.
- (vii) There are infinitely many $n \in \mathbb{Z}$ such that $n^2 + 1$ is a prime number.
- (viii) $x^2 > 2$.

Which of these are propositions?

- (i) is a proposition, and it is true.
- (ii) is a proposition, and it is false. (Some cats are black.)
- (iii) is a proposition, and it is false.

(iv) is not a proposition; it is an instruction not a statement.

(v) is not a proposition. It is a matter of opinion, not something that is true or false.

(vi) is a proposition, but it is false. ($\sqrt{2}$ is only approximately 1.414 so they are not equal.)

(vii) is a proposition. No-one knows whether it is true or false, but it is certainly one or the other.

(viii) is not a proposition: we cannot say it is true or false without knowing what x is.

Notice the difference between (iii) and (viii). In (viii), x occurs as a free variable, and the statement becomes a proposition if we substitute a particular value for x . In (iii), the variable x occurs, but it is not a free variable because we have said the condition must hold for all values of x .

The expression $x^2 > 2$ is an example of a **predicate**: it becomes a proposition (so is either true or false) once we specify a value for its free variable x .

We will often represent a proposition by a single letter, such as P or Q , and treat it as a logical variable. So if P is the proposition " $4 + 3 = 7$ " and Q is the proposition "All cats are grey" then P is true and Q is false. Similarly, we will write a predicate with free variable x as $P(x)$, so if $P(x)$ on $\mathbb{R}$ " $x^2 > 2$ " then $P(3)$ is true and $P(1)$ is false.

For a fixed set S , there is a correspondence between predicates on S and subsets of S . For each predicate P on S we have the subset

$$\{s \in S : P(s) \text{ is true}\}$$

consisting of those elements where P is true. (We could just write $\{s \in S : P(s)\}$.) In the other direction, given a subset A , we have the

predicate $s \in A$. It is clear that these operations are mutually inverse.

1.3.2 Quantifiers

Some useful notation is $\forall$ (the universal quantifier) and $\exists$ (the existential quantifier), which mean *for all* and *there exists* respectively.

So

$$\forall x \in \mathbb{R} : x^2 > 2$$

means "for all real numbers x , we have $x^2 > 2$ ", which is a proposition, and it is false: it is just another way of saying (iii) above.

Similarly

$$\exists x \in \mathbb{R} : x^2 > 2$$

means "there exists a real number x with $x^2 > 2$ ". This is a proposition, and it is true. (For example, $x = 3$ would work.)

Notice that when we write $\forall x$ or $\exists x$, we need to indicate what sort of object x is allowed to be (usually by specifying the set it belongs to).

We can form statements involving several quantifiers, but then the order of the quantifiers is important.

■ **Example 1.12** (i) $\forall x \in \mathbb{R} : \exists y \in \mathbb{R} : y > x$.

This says "for every real number x , there exists a real number y such that $y > x$ ".

This is true: if I give you x , you can always find some y with $y > x$, for instance by choosing $y = x + 1$.

(ii) $\exists y \in \mathbb{R} : \forall x \in \mathbb{R} : y > x$.

All we have done is swapped the order of $\forall x \in \mathbb{R}$ and $\exists y \in \mathbb{R}$.

The new statement says "there exists a real number y such that, for every real number x , we have $y > x$ ".

Is there such a y ? It would have to satisfy $y > x$ for every x , so we

would need $y > 1, y > 2, y > 3, \dots$ (taking $x = 1, x = 2, x = 3, \dots$). Whatever y we start with, there will always be some x big enough that $y \leq x$ (for instance, we could take $x = y + 1$, or just $x = y$). So there is no real number y satisfying $y > x$ for *all* x , and the statement is false.

1.3.3 Operations on Propositions

We can combine propositions by logical operations such as **AND** and **OR**. These are denoted by $\wedge$ and $\vee$ respectively. So if P is the proposition "All cats are grey" (which is false) and Q is the proposition "Paris is the capital of France" (which is true), then $P \wedge Q$ is the proposition

"All cats are grey, and Paris is the capital of France"

(which is false, because not all cats *are* grey), and $P \vee Q$ is the proposition

"All cats are grey, or Paris is the capital of France"

(which is true, because Paris *is* the capital of France).

For any propositions P, Q , the proposition $P \wedge Q$ is true when both of P, Q are true. It is false in all other cases. A convenient way to record this is with a **Truth Table**; we write T for true and F for false.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Similarly, $P \vee Q$ is true if P is true or Q is true (or both: we are using *or* in the inclusive sense.) The truth table is

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

There is also the negation operator $\neg$ (**NOT**), which operates on a single statement. The statement $\neg P$ is true if P is false, and vice versa.

P	$\neg P$
T	F
F	T

1.3.4 Identities for compound propositions

We can use the logical operations to build up compound propositions such as $P \wedge (Q \vee R)$. This proposition will have a truth value which depends on the truth values of P , Q and R . Similarly the proposition $(P \vee Q) \wedge (P \vee R)$ has a truth value depending on those of P , Q and R . In fact, these two propositions always have the same truth value, so they are logically equivalent and we have the logical identity

$$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R) \quad (1.1)$$

(This is a distributive rule relating the logical operators $\wedge$ and $\vee$, similar to the distributive rule $x(y + z) = xy + xz$ relating multiplication and addition of numbers.)

How do we know that the truth values of the two sides (1.1) really are the same? Well, the left-hand side says that P is true, and also either Q or R is true. The right-hand side says that either P and Q are both true, or P and R are both true. A moment's careful thought should be enough to convince you that these are equivalent. However, a more systematic way to check (1.1) is using Truth Tables.

Since we now have three propositions P, Q, R to consider, and each of them may be true or false, we will need 2^3 rows in the Truth Table.

I will number the columns in the Truth Table for ease of reference.

1	2	3	4	5	6	7	8
P	Q	R	$Q \wedge R$	$P \vee (Q \wedge R)$	$P \vee Q$	$P \vee R$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	F	F
F	F	F	F	F	F	F	F

The columns are filled in successively as follows. Columns 1 to 3 contain all possible combinations of truth values for P, Q and R . Column 4 is obtained by applying $\wedge$ to columns 2 and 3, and then column 5 is obtained by applying $\vee$ to columns 1 and 4. So column 5 contains the truth values for the left-hand side of (1.1). Column 6 is obtained by applying $\vee$ to columns 1 and 2. Column 7 is obtained by applying $\vee$ to columns 1 and 3. Finally, column 8 is obtained by applying $\wedge$ to columns 6 and 7. So column 8 contains the truth values for the right-hand side of (1.1). As columns 5 and 8 agree in every row, the two sides of (1.1) agree for every possible combination of truth values of P, Q and R , so the logical identity (1.1) is valid.

Using this, and the correspondence between predicates and subsets, we will revisit Example 1.11.

Let A, B, C be subsets of some set S , and let $P(s), Q(s), R(s)$ be the predicates $s \in A, s \in B, s \in C$ respectively. Then

$$B \cap C = \{s \in S : Q(s) \wedge R(s)\},$$

$$A \cup (B \cap C) = \{s \in S : P(s) \vee (Q(s) \wedge R(s))\}.$$

Also

$$A \cup B = \{s \in S : P(s) \vee Q(s)\},$$

$$A \cup C = \{s \in S : P(s) \vee R(s)\},$$

$$(A \cup B) \cap (A \cup C) = \{s \in S : (P(s) \vee Q(s)) \wedge (P(s) \vee R(s))\}.$$

But for each $s \in S$ we have

$$P(s) \vee (Q(s) \wedge R(s)) = (P(s) \vee Q(s)) \wedge (P(s) \vee R(s))$$

from (1.1), so that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Remark 1.13

Notice how the symbols used for set-theoretic and logical operations match up: the two symbols $\cap$ and $\wedge$ pointing upwards correspond, and the two symbols $\cup$ and $\vee$ pointing downwards correspond. Also the two rounded symbols ($\cap$ and $\cup$) denote operations on sets, and the two straight symbols ($\wedge$ and $\vee$) denote operations on statements.

1.3.5 De Morgan's Laws

When we negate a statement such as $P \wedge Q$ or $P \vee Q$, we negate the two pieces P , Q and also swap $\wedge$ and $\vee$. These are captured in the two logical identities called De Morgan's Laws:

$$\neg(P \wedge Q) = (\neg P) \vee (\neg Q). \quad (1.2)$$

$$\neg(P \vee Q) = (\neg P) \wedge (\neg Q). \quad (1.3)$$

We can check that these are indeed logical identities by writing out the appropriate truth tables. (Do this!) You might have come across another version of De Morgan's Laws for unions and intersections of sets.

To get some insight into what De Morgan's Laws mean, again let P be the proposition *All cats are grey* and let Q be the proposition *Paris is the capital of France*. So $P \wedge Q$ is the proposition

*All cats are grey, **and** Paris is the capital of France.*

What would it mean to say that this sentence is not true? Precisely that

*(Either) not all cats are grey, **or** Paris is not the capital of France.*

This illustrates (1.2). Similarly, to say that the sentence

*All cats are grey, **or** Paris is the capital of France*

is not true is to say that

*Not all cats are grey, **and** Paris is not the capital of France*

which illustrates (1.3).

Here are some more mathematical examples.

■ **Example 1.14** If $P(x)$ is the predicate $x > 2$ and $Q(x)$ is the predicate $x < 100$ then

$$P(x) \vee Q(y)$$

is the predicate $x > 2$ or $y < 100$. (This is a predicate with two free variables x and y .) Its negation $\neg(P(x) \vee Q(y))$ is $(\neg P(x)) \wedge (\neg Q(y))$,

that is

$$x \leq 2 \text{ and } y \geq 100.$$

■ **Example 1.15** What is the negation of $3 < x < 5$? The given statement says that two inequalities are both true: $3 < x$ and $x < 5$. So we need to negate each inequality separately but replace *and* by *or*. The negation is:

$$3 \geq x \text{ or } x \geq 5,$$

which would normally write

$$x \leq 3 \text{ or } x \geq 5.$$

1.3.6 Negating Quantified Statements

Consider the statement

$$\exists x \in \mathbb{Q} : x^2 = 2$$

(which is false: there is no *rational* square root of 2). To say that this statement is false is to say that, whatever rational number x we take, $x^2 \neq 2$, that is

$$\forall x \in \mathbb{Q} : x^2 \neq 2.$$

In general, if we have a statement of the form $\exists x \in S : P(x)$ for some set S and some predicate on S , then its negation will be $\forall x \in S : \neg P(x)$. Thus

$$\neg(\exists x \in S : P(x)) \text{ is equivalent to } \forall x \in S : \neg P(x).$$

In words, to say that it is *not* the case that there is some $x \in S$ with property P is to say that every $x \in S$ fails to have property P . Notice that the condition $x \in S$ does not change: both the original statement and its negation say something about elements of the set S .

Similarly, the negation of $\forall x \in S : P(x)$ is $\exists x \in S : \neg P(x)$, since to say that it is *not* the case that every $x \in S$ has property P is to say that some $x \in S$ fails to have property P . Thus

$$\neg(\forall x \in S : P(x)) \text{ is equivalent to } \exists x \in S : \neg P(x).$$

If we have statements with multiple quantifiers, we can apply these rules step by step.

■ **Example 1.16** Consider the statement $\forall x \in \mathbb{R} : \exists y \in \mathbb{R} : x + y = 2$.

This is a true statement: whatever x we start with, we can find y (depending on x) such that $x + y = 2$. Indeed $y = 2 - x$ will work (and in fact this is the only possible choice of y).

Let us now negate this statement: we want to simplify

$$\neg(\forall x \in \mathbb{R} : \exists y \in \mathbb{R} : x + y = 2).$$

Negating the $\forall$, this is the same as $\exists x \in \mathbb{R} : \neg(\exists y \in \mathbb{R} : x + y = 2)$.

Negating the $\exists$, this is the same as $\exists x \in \mathbb{R} : \forall y \in \mathbb{R} : \neg(x + y = 2)$.

Finally, negating the $=$, we get $\exists x \in \mathbb{R} : \forall y \in \mathbb{R} : x + y \neq 2$.

This is a false statement (as we expect, since we negated a true statement). No real number x can have the property that $x + y \neq 2$ for *all* real numbers y , since if we choose $y = 2 - x$ we will certainly have $x + y = 2$.

1.3.7 Implication

We often come across assertions of the form *if P is true then Q is true*.

We have already seen this in relation to proving that a set A is a subset of another set B : we have to show that if $x \in A$ then $x \in B$.

The assertion might be expressed using different words, for example *Q holds whenever P holds*, or *P implies Q* .

We can treat *implies* as a logical operator for combining propo-

sitions, in the same manner as $\wedge$ or $\vee$. We use the symbol $\Rightarrow$, so $P \Rightarrow Q$ means P *implies* Q . However, doing so means that we must interpret $P \Rightarrow Q$ as having a truth value that only depends on the truth values of P and Q . This does not always match exactly with the way we use phrases like *if ... then* in everyday life, so we need to be careful.

Consider the statement

If you behave then you will get an ice cream.

As well as being a promise of a reward for good behaviour, it might be understood as a threat concerning the consequences of bad behaviour. However, the statement does *not actually state* anything about what happens if you do not behave. In that case, you might get an ice cream or you might not. The only circumstances in which the statement is a lie is if you *do* behave but *don't* get an ice-cream.

As another example, what about the following statement:

If it is raining in Nottingham then Paris is the capital of France?

Is this true or false? Let us consider the two possibilities. Suppose it is raining in Nottingham. Then it is indeed the case that Paris is the capital of France. So that case is OK. Suppose it is not raining in Nottingham. Then our statement does not claim anything about capital of France, so that case is also OK, whether Paris is the capital of France or not. We should therefore count our statement as being true. The only situation in which we could declare it to be false would if it *was* raining in Nottingham, and Paris was *not* the capital of France.

In general, the statement $P \Rightarrow Q$ only says that if P is true then Q must be true as well. If P is false, it does not say anything about

Q , and if Q is true, it does not tell us anything about P . The only situation in which we can say that $P \Rightarrow Q$ is false is when P is true but Q is false. So the Truth Table for $P \Rightarrow Q$ is

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Remark 1.17

- $P \Rightarrow Q$ does not necessarily mean there is any *causal* relationship between P and Q : the weather in Nottingham does not *make* Paris the capital of France.
- $P \Rightarrow Q$ is not all the same thing as $Q \Rightarrow P$: the statement we considered definitely does not say

If Paris is the capital of France then it is raining in Nottingham.

The logical operation $\Rightarrow$ can be built up from our other logical operations: we have the logical identity

$$P \Rightarrow Q = (\neg P) \vee Q.$$

Check this by writing out the Truth Table for $(\neg P) \vee Q$.

1.3.8 Deduction

In constructing logically correct arguments (such as mathematical proofs), a basic step is *deduce* that a certain conclusion necessarily follows from given *premises*. The point is not whether either the premises or the conclusion are *in fact* true, but the logical relation

between them. For example, if we start with the premises that

Anything that has wings can fly

and

Percy has wings

then we can deduce from these premises that

Percy can fly.

It is not relevant to point out that Percy might be a penguin (or a palace).

On the other hand, if we have the premises

All dogs have four legs

and

Fido has four legs

we cannot deduce that

Fido is a dog.

(The premises do not tell us that *only* dogs have four legs. Perhaps Fido is a horse.)

What has this got to do with Propositional Calculus? If P , Q , ... are propositions and we are given the premises $P \Rightarrow Q$ and P , then we can deduce Q . (To say this another way, if we know that the propositions $P \Rightarrow Q$ and P are both true, we know that Q must be true as well.) We can see this from the truth table:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

In each row with T for both P and $P \Rightarrow Q$ (in fact, there is only one such row), we also have T for Q . Thus whenever P and $P \Rightarrow Q$ are true then Q is necessarily true as well.

Similarly, if we are given the premise $P \wedge Q$, we can deduce P . (We can also deduce Q .)

From the premise $P \vee Q$, we cannot deduce P . (It might be that Q is true but P is false). However, from the premise P we can deduce $P \vee Q$.

1.4 General Methods of Proof

We will look briefly some logically correct patterns that a proof might follow. We won't discuss Proof by Induction here, since that is only applicable to statements depending on the natural numbers.

1.4.1 Direct Proof

If we want to prove a statement of the form $\forall x \in S : P(x) \Rightarrow Q(x)$ then the most obvious approach is:

Let $x \in S$ and assume $P(x)$ is true.

Deduce that $Q(x)$ is true.

■ **Example 1.18** *Prove that for every natural number n , if n is even then n^2 is even.*

Here S is the set $\mathbb{N}$ of natural numbers, $P(n)$ is the predicate n is even and $Q(n)$ is the predicate n^2 is even.

Proof. Let $n \in \mathbb{N}$ with n even.

Then $n = 2m$ for some $m \in \mathbb{Z}$.

So $n^2 = (2m)^2 = 2(2m^2)$.

Since $2m^2 \in \mathbb{Z}$, this shows that n^2 is even. ■

1.4.2 Contrapositive Proof

The **contrapositive** of the implication $P \Rightarrow Q$ is $(\neg Q) \Rightarrow (\neg P)$. The propositions $P \Rightarrow Q$ and $(\neg Q) \Rightarrow (\neg P)$ are logically equivalent. (If that is not obvious to you, check it by writing out their truth tables.) So if we want to prove $\forall x \in S : P(x) \Rightarrow Q(x)$, we could instead prove that $\forall x \in S : \neg Q(x) \Rightarrow \neg P(x)$.

■ **Example 1.19** *Prove that for every natural number n , if n^2 is even then n is even.*

(This is the **converse** of the statement in the previous example.)

Here S is the set $\mathbb{N}$ of natural numbers, $P(n)$ is the predicate n^2 is even and $Q(n)$ is the predicate n is even. In this case, a direct proof is not straightforward: if $n^2 = 2m$ with $m \in \mathbb{Z}$, we cannot easily see that there is an integer k with $n = 2k$. So instead we prove the contrapositive statement:

For every natural number n , if n is odd then n^2 is odd.

Proof. Let $n \in \mathbb{N}$ and suppose that n is odd. [Here we are assuming $\neg Q(n)$.]

Then $n = 2m + 1$ for some $m \in \mathbb{Z}$.

So $n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 2k$ where $k = 2m^2 + 2m \in \mathbb{Z}$.

Hence n^2 is odd. ■

1.4.3 Proof by Contradiction

One way to prove that a statement is *true* is to assume that it is *false* and deduce a contradiction (i.e. something that we know is definitely false). In terms of Propositional Calculus, we are using that from $(\neg P) \Rightarrow F$ we can deduce P . (Check this with a Truth Table!)

Here is an ancient and famous example.

■ **Example 1.20** Prove that there is no $x \in \mathbb{Q}$ with $x^2 = 2$. (i.e. Prove that $\sqrt{2}$ is irrational.)

Proof. Suppose for a contradiction that there is some $x \in \mathbb{Q}$ with $x^2 = 2$.

We can assume $x > 0$ (otherwise replace it by $-x$). Since x is rational, we have $x = m/n$ for some $m, n \in \mathbb{N}$. Moreover, we can assume that the fraction is in lowest terms, so m, n have no common factors > 1 .

Then $(m/n)^2 = x^2 = 2$, so $m^2 = 2n^2$.

Hence m^2 is even. By Example 1.19, m is also even, say $m = 2k$ with $k \in \mathbb{N}$.

Then $(2k)^2 = 2n^2$, so $2k^2 = n^2$. So n^2 is even, and by Example 1.19 again, n is even, say $n = 2h$ for $h \in \mathbb{N}$.

We have shown that m and n are both even, so they have 2 as a common factor. This is a contradiction, since we chose the fraction m/n so that m and n have no common factor > 1 .

Hence our original assumption cannot be true: there is no $x \in \mathbb{Q}$ with $x^2 = 2$. ■

This proof only uses properties of odd and even numbers, and does not rely on any general facts about prime factorisations. You might like to think about what exactly you would need to use in order to prove that (for example) $\sqrt{11}$ is irrational.

2 FUNCTIONS

2.1 What is a function?

When you think about functions, you probably think of something like $\sin x$ or $\log x$ which gives an output (a real number) for each input x (a real number, possibly with some restrictions, for example, $\log x$ is only defined for $x > 0$).

We now consider functions in a more general way: the inputs can come from any set A , and the outputs can be in any set B . We therefore need to say what these sets are in order to specify the function.

Definition 2.1 A **function** f from a set A to a set B is a rule which associates to *every* element $a \in A$ one and only one element $f(a)$ in B .

We write $f: A \rightarrow B$ to indicate that f is a function from A to B . The set A is called the **domain** of the function, and B is called its **codomain**.

There are several things to note about this definition. Firstly, to specify a function properly, we need to specify its domain and codomain (the sets A and B). Secondly, we need to specify $f(a)$ for *every* element a without exception (not just most elements, or a “typical” element). Thirdly, for each a , there must not be any ambiguity about the element $f(a)$: there must be one and only one value which works. Fourthly, a function need not be given by a formula; it could be given by a table of values, or by splitting cases, or in some other way.

Strictly speaking, we should not call $\sin(x)$ a function. Instead we have a function $\sin: \mathbb{R} \rightarrow \mathbb{R}$ which takes each $x \in \mathbb{R}$ to the value $\sin x \in \mathbb{R}$. So $\sin x$ is the *value* of the function at x . However, mathematicians are often a little bit sloppy about this sort of thing. We can also say we have a function $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = \sin x$

for each x .

Remark 2.2

Our conventions in this module are slightly different from those in MTH1002 Mathematical Methods. In MTH1002, all the functions considered are real-valued functions of a real number, so the domain and codomain are always subsets of $\mathbb{R}$. If a function f is given by a formula and the domain is not explicitly stated, then the domain is assumed to be the *maximal domain* for the formula, i.e. the largest subset of $\mathbb{R}$ on which the formula can be evaluated. The codomain is often implicitly taken to be the whole of $\mathbb{R}$, or sometimes the range of f . In this module, we want to consider functions whose domain and codomain could be any sets (not necessarily subsets of $\mathbb{R}$), so the domain and the codomain *will* always be stated explicitly, and are taken to be part of the definition of the function.

Here are some examples which do, or don't, fulfil all the conditions to define a function.

■ **Example 2.3** $f: \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = x^3 + e^x$ for all $x \in \mathbb{R}$, is a function. (For each x , the formula unambiguously gives a value $f(x) \in \mathbb{R}$.) This is typical of the sort of functions we usually think about in calculus, where both the domain and the codomain are (subsets of) the set of real numbers, and the function has a well-behaved graph (i.e. the function is continuous and differentiable, so its graph is a smooth curve).

■ **Example 2.4** $f: \mathbb{R} \rightarrow \mathbb{N}$, given by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 2 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

This **is** a function, since it gives an unambiguous value $f(x) \in \mathbb{N}$ for each $x \in \mathbb{R}$, even though it doesn't make much sense to look at the graph of this function. (The function is nowhere continuous.)

■ **Example 2.5** $f: \mathbb{R} \rightarrow \mathbb{R}$, given by

$$f(x) = \frac{1}{x}$$

does not define a function, since $f(0)$ is not defined. We can fix this by modifying the domain. Write

$$\mathbb{R} \setminus \{0\} = \{x \in \mathbb{R} : x \neq 0\},$$

the set of *non-zero* real numbers. (This is also sometimes written $\mathbb{R}^\times$.)

Then $f_1(x) = 1/x$ defines a function $f_1: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$. Alternatively, we can include the bad point $x = 0$ by defining the value of the function separately there:

$$f_2: \mathbb{R} \rightarrow \mathbb{R}, \quad f_2(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0; \\ 0 & \text{if } x = 0; \end{cases}$$

does define a function. However f_1 and f_2 are *different* functions, because they do not have the same domain.

■ **Example 2.6** Let $A = \{1, 2, 3, 4\}$. Then we can define a function $f: A \rightarrow \mathbb{R}$ by the table

a	1	2	3	4
$f(a)$	17	$\sqrt{5}$	π	-2

■ **Example 2.7** $f: \mathbb{Z} \rightarrow \mathbb{Z}$, given by

$$f(n) = \begin{cases} n + 1 & \text{if } n \geq 0, \\ n - 1 & \text{if } n \leq 0 \end{cases}$$

does **not** define a function since $f(0)$ is ambiguously defined; the

first case of the definition says $f(0) = 1$ while the second case says $f(0) = -1$.

■ **Example 2.8** $f: \mathbb{R} \rightarrow \mathbb{Z}$, given by

$$f(x) = \text{“the nearest integer to } x\text{”}$$

appears to define a function, e.g. $f(8/5) = 2$, $f(\pi) = 3$. But what is $f(7/2)$? Is it 3 or 4? Since there is no unique nearest integer in this case, the value of $f(7/2)$ is ambiguous, so we **have not** defined a function.

■ **Example 2.9** $f: \mathbb{R} \rightarrow \mathbb{Z}$, given by

$$f(x) = \text{“the largest integer } \leq x\text{”}$$

does define a function, e.g. $f(3/4) = 0$, $f(-7/3) = -3$, $f(7) = 7$.

We count two functions as equal if they have the same domain and the same codomain, and agree at all elements of the domain:

Definition 2.10 Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be functions. Then f and g are equal (written $f = g$) if $A = C$ and $B = D$ and $f(x) = g(x)$ for all $x \in A$.

■ **Example 2.11** Let

$$f: \mathbb{Z} \rightarrow \mathbb{N}, \quad f(n) = n^2 + 1;$$

$$g: \mathbb{Z} \rightarrow \mathbb{Z}, \quad g(n) = n^2 + 1;$$

$$h: \mathbb{N} \rightarrow \mathbb{N}, \quad h(n) = n^2 + 1;$$

$$k: \mathbb{N} \rightarrow \mathbb{N}, \quad k(n) = (n + 1)^2 - 2n.$$

Then f , g , h , k are all functions. (They take every element n of the specified domain unambiguously to an element of the specified

codomain.) We have $f \neq g$ since the codomains do not match (even though the functions are given by the same formula). Similarly $g \neq h$, but $h = k$.

2.2 Surjective and Injective Functions

For a function $f: A \rightarrow B$, every a in A corresponds to exactly one element $f(a)$ in B . (That is what it means to be a function.) However, if we start with $b \in B$, there may be no $a \in A$ with $f(a) = b$, or there may be more than one such a . For example, if $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^2$, then there is no $x \in \mathbb{R}$ with $f(x) = -1$, but there are two values of x with $f(x) = 4$.

Definition 2.12 For a function $f: A \rightarrow B$:

- (i) We say f is **surjective** (or f is **onto**, or f is a **surjection**) if, for each $b \in B$, there is *at least* one $a \in A$ with $f(a) = b$.
- (ii) We say f is **injective** (or f is **one-to-one**, or f is an **injection**) if, for each $b \in B$, there is *at most* one $a \in A$ with $f(a) = b$. So to say that f is injective means that, for any elements $a, a' \in A$,

$$f(a) = f(a') \Rightarrow a = a'.$$

- (iii) We say f is **bijective** (or f is a **one-to-one correspondence**, or f is a **bijection**) if f is both surjective and injective. Thus f is bijective if, for each $b \in B$, there is *exactly* one $a \in A$ with $f(a) = b$.

■ **Example 2.13** $f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(n) = 2n + 1$.

f is *not* surjective; there is (for example) no $n \in \mathbb{Z}$ with $f(n) = 0$.

f is injective: if $f(n) = f(n')$ then $2n + 1 = 2n' + 1$, which means that $n = n'$.

■ **Example 2.14** $f: \mathbb{R} \rightarrow \mathbb{Z}$, given by

$$f(x) = \text{“the largest integer } \leq x\text{”}.$$

f is surjective: given $n \in \mathbb{Z}$ we have $f(n) = n$.

f is *not* injective: e.g. $f(1/2) = f(0) = 0$.

■ **Example 2.15** $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x(x-1)(x-2)$.

We can easily sketch a graph of this function: the function is continuous, $f(x) = 0$ when $x = 0, 1$ or 2 (and nowhere else). Moreover $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

Given any $b \in \mathbb{R}$, there will be at least one $a \in \mathbb{R}$ with $f(a) = b$. (The graph must cut the horizontal line $y = b$ at least once, since it is continuous and $f(x)$ can take arbitrarily large positive and negative values.) Thus f is surjective.

However, $f(0) = f(1) = f(2) = 0$, so the function is *not* injective. (In terms of the graph, there are some values of b , such as $b = 0$, for which the horizontal line $y = b$ meets the graph more than once.)

Here are some remarks on surjective and injective functions:

- (1) Being surjective and being injective are **not** alternatives; a function may have either one of these properties, or both, or neither.
- (2) We sometimes define the **range** (or **image**) of a function $f: A \rightarrow B$ to be the set of values actually taken by the function:

$$\begin{aligned} \text{Range}(f) &= \{b \in B : \text{there is some } a \in A \text{ with } f(a) = b\} \\ &= \{f(a) : a \in A\}. \end{aligned}$$

So the range of f is a subset of the codomain B , but in general need not be the whole of B . Then f is surjective if and only if its range is equal to its codomain.

- (3) To see if a function is surjective/injective, we need to take into account its domain and codomain.
- (4) Let $f: A \rightarrow B$ be a function.
- (i) To show f is *not* surjective, it is enough to find one particular $b \in B$ which cannot be written as $f(a)$ for any $a \in A$, but
 - (ii) to show f is surjective, we need a general argument to show that *every* $b \in B$ can be written in the form $f(a)$ for some $a \in A$. (This might involve finding a formula for a in terms of b , though there might be more than one a which works.)
 - (iii) To show f is *not* injective, it is enough to find two particular elements $a, a' \in A$ (with $a \neq a'$) such that $f(a) = f(a')$, but
 - (iv) to show f is injective, we need a general argument to show that whenever $f(a) = f(a')$ (with $a, a' \in A$), we must have $a = a'$.
 - (v) To show f is bijective, we must show separately that it is surjective and it is injective. Alternatively, we can show f has a (two-sided) inverse function – see later.

Here are some further examples:

■ **Example 2.16** $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^3$.

For each $b \in \mathbb{R}$, we can take $x = \sqrt[3]{b}$. (This is positive if b is positive and negative if b is negative). Then $f(x) = b$. This shows that b is surjective. On the other hand, as $f(x)$ is strictly increasing (if $x < x'$ then $f(x) < f(x')$), there can only be one value of x with $f(x) = b$, so f is injective. Hence f is a bijection.

(Since f is a continuous function from the real numbers to the real numbers, we can interpret this in terms of the graph: for each b , the horizontal line $y = b$ meets the graph in *exactly* one point.)

■ **Example 2.17** The function $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \sin x$ is neither surjective nor injective. It is not surjective since there is no $x \in \mathbb{R}$ with $\sin(x) = 2$. It is not injective since $\sin(0) = \sin(\pi) = 0$. We can however modify f to get a function which is both surjective and injective (i.e., which is bijective) by reducing its domain and codomain. Let

$$A = [-\pi/2, \pi/2] = \{x \in \mathbb{R} : -\pi/2 \leq x \leq \pi/2\}$$

and

$$B = [-1, 1] = \{y \in \mathbb{R} : -1 \leq y \leq 1\}.$$

Then we can define a function $\hat{f}: A \rightarrow B$ by $\hat{f}(x) = \sin x$ for all $x \in A$. This makes sense since $-1 \leq \sin(x) \leq 1$ for all $x \in A$ (indeed, for all $x \in \mathbb{R}$), so that $\hat{f}(x)$ really is an element of B . This new function is both surjective and injective.

2.3 The “algebra” of functions

2.3.1 Composition of functions

We can **compose** functions $f: A \rightarrow B$ and $g: B \rightarrow C$ to form a new function $g \circ f: A \rightarrow C$ (the **composite** of f and g), defined by

$$g \circ f(a) = g(f(a)) \text{ for all } a \in A.$$

There are two things to note here:

(1) The set B appears twice! The composite $g \circ f$ is only defined when the domain of g coincides with (or, at least, contains) the codomain of f .

(2) $g \circ f$ means “do f first, then g ”. (For example, to evaluate $\log \sin x$, i.e. to evaluate the composite $\log \circ \sin$ at x , you take x , first find its sine, and then take the log of that.) This comes about because of the convention that we write the function before (to the left of) its argument, i.e. we write $f(x)$ rather than $(x)f$.

Remark 2.18

Since in this module we always specify the domain and codomain of a function explicitly, our description of composition of functions is a bit different to that in MTH1002 (see Remark 2.2). In MTH1002, if $f : A \rightarrow \mathbb{R}$ and $g : B \rightarrow \mathbb{R}$ are functions given by formulae, and A, B are the maximal domains of these formulae, then the maximal domain of $h = g \circ f$ is the set $A_0 := \{a \in A : f(a) \in B\}$. In the conventions of this module, the function h is obtained as follows. We first replace f by a new function $\bar{f} : A_0 \rightarrow B$ with smaller domain and codomain than f , where $\bar{f}$ is defined by $\bar{f}(a) = f(a)$ for all $a \in A_0$. This makes sense since $\text{Range}(\bar{f}) \subseteq B$. We then compose $\bar{f}$ with g to get $h = g \circ \bar{f}$.

■ **Example 2.19** Let $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = 5x, \quad g(x) = x - 2, \quad h(x) = x^2.$$

Then

$$f \circ g(x) = f(x - 2) = 5x - 10,$$

whereas

$$g \circ f(x) = g(5x) = 5x - 2.$$

This shows that the order of composition matters: $f \circ g$ need not be the same as $g \circ f$ (even when both make sense).

On the other hand, if we compose $(f \circ g)$ with h we get

$$(f \circ g) \circ h(x) = (f \circ g)(x^2) = 5x^2 - 10,$$

and, as $g \circ h(x) = x^2 - 2$, we also have

$$f \circ (g \circ h)(x) = f(x^2 - 2) = 5x^2 - 10.$$

Thus (at least in this example)

$$(f \circ g) \circ h = f \circ (g \circ h) \quad (\text{as functions}).$$

We will see that this always happens.

Proposition 2.20 (Associative property of function composition) Suppose we have functions $h: A \rightarrow B$, $g: B \rightarrow C$, $f: C \rightarrow D$ (for any sets A, B, C, D). Then all the functions

$$f \circ g: B \rightarrow D, \quad (f \circ g) \circ h: A \rightarrow D,$$

$$g \circ h: A \rightarrow C, \quad f \circ (g \circ h): A \rightarrow D$$

make sense (i.e. the domains and codomains match up so that the compositions are defined), and

$$(f \circ g) \circ h = f \circ (g \circ h).$$

(This means we can simply write $f \circ g \circ h$ without ambiguity.)

Proof. For each $a \in A$ we have (just using the definition of function composition)

$$f \circ (g \circ h)(a) = f(g \circ h(a)) = f(g(h(a)))$$

and also

$$(f \circ g) \circ h(a) = (f \circ g)(h(a)) = f(g(h(a))).$$

Thus

$$f \circ (g \circ h)(a) = (f \circ g) \circ h(a) \text{ for each } a \in A,$$

so that

$$(f \circ g) \circ h = f \circ (g \circ h).$$



2.3.2 Identity Functions

Definition 2.21 For any set A , the **identity function** on A is the function $\text{id}_A: A \rightarrow A$ given by

$$\text{id}_A(a) = a \text{ for all } a \in A,$$

Composing with an identity function doesn't change anything:

Proposition 2.22 For any function $f: A \rightarrow B$,

$$f \circ \text{id}_A = f \text{ and } \text{id}_B \circ f = f.$$

(Note that we have to take the identity function on the correct set for the composition to make sense.)

Proof. For each $a \in A$, we have

$$f \circ \text{id}_A(a) = f(\text{id}_A(a)) = f(a)$$

so $f \circ \text{id}_A = f$.

Also, for each $f \in A$, we have

$$\text{id}_B \circ f(a) = \text{id}_B(f(a)) = f(a)$$

so $\text{id}_B \circ f = f$.



2.3.3 Inverse functions

Definition 2.23 An inverse (function) to a function $f: A \rightarrow B$ is a function $g: B \rightarrow A$ such that

$$g \circ f = \text{id}_A \text{ and } f \circ g = \text{id}_B.$$

So an inverse function (without any further qualification) means a *two-sided* inverse function. It is clear from the definition that if g is an inverse to f then f is an inverse to g .

■ **Example 2.24** The function $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = (3x - 2)/5$ has inverse function $g: \mathbb{R} \rightarrow \mathbb{R}$ with $g(y) = (5y + 2)/3$.

To see this, we check that for all $x \in \mathbb{R}$,

$$g \circ f(x) = g\left(\frac{3x - 2}{5}\right) = \frac{1}{3} \left(5 \left(\frac{3x - 2}{5}\right) + 2\right) = x,$$

so that $g \circ f = \text{id}_{\mathbb{R}}$, and for all $y \in \mathbb{R}$,

$$f \circ g(y) = f\left(\frac{5y + 2}{3}\right) = \frac{1}{5} \left(3 \left(\frac{5y + 2}{3}\right) - 2\right) = y,$$

so that $f \circ g = \text{id}_{\mathbb{R}}$.

■ **Example 2.25** Let $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$, the set of (strictly) positive real numbers. Then the function $\ln: \mathbb{R}^+ \rightarrow \mathbb{R}$ has inverse function $\exp: \mathbb{R} \rightarrow \mathbb{R}^+$, since

(i) for each $x \in \mathbb{R}^+$, we have $\exp(\ln(x)) = x$, so $\exp \circ \ln = \text{id}_{\mathbb{R}^+}$,

and

(ii) for each $y \in \mathbb{R}$, we have $\ln(\exp(y)) = y$, so $\ln \circ \exp = \text{id}_{\mathbb{R}}$.

Not every function has an inverse, but a function cannot have more than one inverse. We can see this using the formal properties of function composition and identity functions given in Propositions 2.20 and 2.22.

Proposition 2.26 Uniqueness of inverse functions If a function $f: A \rightarrow B$ has an inverse then this inverse is unique.

Proof. Suppose that both of the functions $g: B \rightarrow A$ and $g': B \rightarrow A$ are inverses to f . We must show that $g = g'$.

Since g is an inverse to f , we know that $g \circ f = \text{id}_A$ (and also $f \circ g = \text{id}_B$).

Since g' is an inverse to f , we know that $f \circ g' = \text{id}_B$ (and also $g' \circ f = \text{id}_A$).

So

$$\begin{aligned}
 g' &= \text{id}_A \circ g' && \text{(Proposition 2)} \\
 &= (g \circ f) \circ g' \\
 &= g \circ (f \circ g') && \text{(Proposition 1)} \\
 &= g \circ (\text{id}_B) \\
 &= g && \text{(Proposition 2).}
 \end{aligned}$$

■

The next result tells us exactly when a function does have an inverse.

Theorem 2.27 A function $f: A \rightarrow B$ has an inverse function if and only if f is a bijection.

Proof. We show the two directions of the “if and only if” separately.

(1) Suppose that f has inverse $g: B \rightarrow A$. Then $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. We need to show that f is a bijection, i.e., it is both surjective and injective.

To see f is surjective, let $b \in B$. Let $a = g(b)$. Then $f(a) = f(g(b)) = b$, since $f \circ g = \text{id}_B$. So we have found $a \in A$ with $f(a) = b$. This shows that f is surjective.

To see f is injective, suppose that $f(a_1) = f(a_2)$ for some $a_1, a_2 \in A$. We need to show that $a_1 = a_2$. But applying g , we have $g(f(a_1)) = g(f(a_2))$, so that $a_1 = a_2$ since $g \circ f = \text{id}_A$. This shows that f is injective.

We have now shown that if f has an inverse then f is a bijection.

(2) Now suppose that f is a bijection. We need to define a suitable function $g: B \rightarrow A$ and check that it really is an inverse to f .

For each $b \in B$, there is exactly one $a \in A$ with $f(a) = b$. Call

this $g(b)$. Since there is no ambiguity about the choice of $g(b)$, this defines a function $g: B \rightarrow A$.

Starting with $b \in B$, we have $g(b) = a$ where $f(a) = b$, so that $f(g(b)) = f(a) = b$. This shows that $f \circ g = \text{id}_B$.

Starting with $a \in A$, let $b = f(a)$, and let $a' = g(b)$. Then, by the definition of g , we know that a' is the unique element of a with $f(a') = b$. But $f(a) = b$ (that is how we defined b), so $a' = a$ and $g(f(a)) = g(b) = a' = a$. This shows that $g \circ f = \text{id}_A$.

Hence g is an inverse to f . ■

Definition 2.28 Let $f: A \rightarrow B$ be a bijection. Then we write f^{-1} for its inverse function $f^{-1}: B \rightarrow A$.

This definition is valid (i.e. it does unambiguously tell us the meaning of f^{-1} because Theorem 2.27 tells us that f does indeed have one and only one inverse function.

■ **Example 2.29** Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be given by

$$f(n) = \begin{cases} n + 1 & \text{if } n \text{ is odd,} \\ n - 1 & \text{if } n \text{ is even.} \end{cases}$$

We will show that f is a bijection and find its inverse function.

First note that if n is odd then $f(n)$ is even, and vice versa. Thus if $f(n)$ is odd then n must be even, and if $f(n)$ is even then n must be odd.

To see that f is bijective, we show separately that f is surjective and f is injective.

To show that f is surjective, let $m \in \mathbb{Z}$. Then either m is odd or m is even. If m is odd, set $n = m + 1$. Then n is even and $f(n) = n - 1 = m$. If m is even, set $n = m - 1$. Then n is odd, and $f(n) = n + 1 = m$. Either way, we have found an $n \in \mathbb{Z}$ with $f(n) = m$, so f is surjective.

To show that f is injective, let $n_1, n_2 \in \mathbb{Z}$ with $f(n_1) = f(n_2)$. Then either $f(n_1)$ is odd or $f(n_1)$ is even. If $f(n_1) = f(n_2)$ is odd, then n_1 and n_2 are both even, and we have $n_1 - 1 = n_2 - 1$, so that $n_1 = n_2$. If $f(n_1) = f(n_2)$ is even, then n_1 and n_2 are both odd, and we have $n_1 + 1 = n_2 + 1$, so that again $n_1 = n_2$. Either way, $n_1 = n_2$, so f is injective.

Since f is both surjective and injective, it is bijective.

The argument showing that g is surjective tells us that the only candidate for f^{-1} is the function $g : \mathbb{Z} \rightarrow \mathbb{Z}$ where

$$g(m) = \begin{cases} m + 1 & \text{if } m \text{ is odd,} \\ m - 1 & \text{if } m \text{ is even.} \end{cases}$$

But this is exactly the function f , so f is in fact its own inverse:

$$f \circ f = \text{id}_{\mathbb{Z}}.$$

Proposition 2.30 (The inverse of a composite of bijections) Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections. Then $g \circ f : A \rightarrow C$ is also a bijection, and its inverse is

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

(Note that the order is reversed.)

Proof. We check that the function $f^{-1} \circ g^{-1} : C \rightarrow A$ is an inverse to $g \circ f$, using the associative property of function composition:

$$\begin{aligned} (g \circ f) \circ (f^{-1} \circ g^{-1}) &= g \circ (f \circ (f^{-1} \circ g^{-1})) \\ &= g \circ ((f \circ f^{-1}) \circ g^{-1}) \\ &= g \circ (\text{id}_B \circ g^{-1}) \\ &= g \circ g^{-1} = \text{id}_C. \end{aligned}$$

Similarly $(f^{-1} \circ g^{-1}) \circ (g \circ f) = \text{id}_A$. So $g \circ f$ has inverse $f^{-1} \circ g^{-1}$.

Since $g \circ f$ has an inverse, it must be a bijection (by the Theorem). ■

Definition 2.31 (One-sided inverses)

For a function $f: A \rightarrow B$, we say:

- (i) $g: B \rightarrow A$ is a *left inverse* to f if $g \circ f = \text{id}_A$;
- (ii) $h: B \rightarrow A$ is a *right inverse* to f if $f \circ h = \text{id}_B$.

One-sided inverses (i.e. left inverses or right inverses) need not be unique, and a function can have a one-sided inverse without being a bijection.

■ **Example 2.32** Let $f: \mathbb{Z} \rightarrow \mathbb{R}$ with $f(x) = x$, so f is the inclusion function of $\mathbb{Z}$ into $\mathbb{R}$, and let $g: \mathbb{R} \rightarrow \mathbb{Z}$ be the function given by

$$g(x) = \begin{cases} x & \text{if } x \in \mathbb{Z}, \\ 0 & \text{if } x \notin \mathbb{Z}. \end{cases}$$

Then $g \circ f = \text{id}_{\mathbb{Z}}$, but $f \circ g \neq \text{id}_{\mathbb{R}}$, e.g. $f \circ g(\pi) = f(0) = 0 \neq \pi$.

Thus g is a left inverse (but not a right inverse) to f , and f is a right inverse (but not a left inverse) to g . Note that f is injective but not surjective, and g is surjective but not injective.

Remark 2.33**A comment on inverse trigonometric functions**

The function $\sin: \mathbb{R} \rightarrow \mathbb{R}$ is neither surjective nor injective, so is certainly not a bijection. According to Theorem 2.27, it therefore cannot have an inverse function. So what do we mean by $\sin^{-1}$?

We can restrict both the domain and codomain of $\sin$ to get a function

$$\widehat{\sin}: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

which is a bijection. Here the notation $[a, b]$ means the closed interval from a to b :

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

This function $\widehat{\sin}$ does have an inverse

$$\widehat{\sin}^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

and that is what the $\sin^{-1}$ button on your calculator corresponds to.

Similar comments apply to $\cos^{-1}$ and $\tan^{-1}$.

2.4 Permutations

A permutation is a special type of function. Here is a (general and abstract) definition.

Definition 2.34 Let X be any set. Then a *permutation* on X is a bijection $\pi: X \rightarrow X$.

We can immediately record several properties of permutations that follow from the Algebra of Functions in the previous subsection.

- (1) If σ, τ are permutations on X , so is $\sigma \circ \tau$.
- (2) For any permutations σ, τ, π on X , the permutations $(\sigma \circ \tau) \circ \pi$ and $\sigma \circ (\tau \circ \pi)$ are equal.
- (3) The identity function $\text{id}_X: X \rightarrow X$ is a permutation on X .
- (4) Each permutation σ on X has an inverse function $\sigma^{-1}: X \rightarrow X$, so that $\sigma \circ \sigma^{-1} = \text{id}_X$ and $\sigma^{-1} \circ \sigma = \text{id}_X$. The function σ^{-1} is itself a permutation. (It is a bijection since it has an inverse function, namely σ itself.)

We will see later that a set with these four properties is a group. So the collection (set) of all permutations on X is a group under the operation of composing permutations.

We now suppose that X is a finite set, say with n elements. By labelling the elements of X as $1, 2, \dots, n$, and identifying each element with its label, there is no harm in supposing that $X = \{1, 2, \dots, n\}$.

Definition 2.35 For any natural number n , we write S_n for the set of all permutations on the set $\{1, 2, \dots, n\}$. (It is sometimes written $\mathfrak{S}_n$.)

■ **Example 2.36** We take $n = 3$, and consider some functions from the set $\{1, 2, 3\}$ to itself.

i	1	2	3
$\sigma(i)$	2	3	1
$\tau(i)$	1	3	2
$\text{id}(i)$	1	2	3
$\alpha(i)$	2	1	2

The functions σ and τ , as specified in the table, are both permutations: σ is injective because the 3 entries in the row labelled $\sigma(i)$ are all different, and it is surjective since each element in the set $\{1, 2, 3\}$

occurs somewhere in that row. Thus σ is a bijection, so it is a permutation on $\{1, 2, 3\}$. Similarly, τ is a permutation. The identity function id on X is also a permutation. But the function α is not a permutation: indeed, it is neither injective nor surjective. (It is not injective since $\alpha(1) = \alpha(3) = 2$, and it is not surjective since there is no i with $\alpha(i) = 3$.)

Thus $\sigma \in S_3$ and $\tau \in S_3$ but $\alpha \notin S_3$.

Lemma 2.37 $|S_n| = n!$. In other words, there are exactly $n!$ permutations on $\{1, 2, \dots, n\}$.

Proof. Choosing a permutation σ in S_n amounts to filling in the n boxes in the second row of the table

i	1	2	3	$\dots$	n
$\sigma(i)$				$\dots$	

There are n possibilities for $\sigma(1)$.

There are $n - 1$ possibilities for $\sigma(2)$, as we cannot reuse $\sigma(1)$ since σ must be injective.

There are $n - 2$ possibilities for $\sigma(3)$, as we cannot reuse $\sigma(1)$ or $\sigma(2)$.

Carrying on in this way, we end with 1 possibility for $\sigma(n)$.

We have then completed the table and produced an injective function σ . But all n possible values have been used, so σ is also surjective. Hence it is a permutation.

The number of permutations is therefore $n \times (n - 1) \times (n - 2) \times \dots \times 1 = n!$. ■

When we compose permutations, we will usually omit the symbol $\circ$, and write $\sigma\tau$ instead of $\sigma \circ \tau$. We will write σ^2 for $\sigma\sigma$, σ^3 for $\sigma\sigma\sigma$, etc, and we will adopt the convention $\sigma^0 = \text{id}$. Then we have the usual rule of exponents $\sigma^k \sigma^m = \sigma^{k+m}$.

2.4.1 Cycle notation

This is a more efficient way of writing permutations than writing them as a table. Let us consider the permutation $\pi \in S_8$ given by the table

i	1	2	3	4	5	6	7	8
$\pi(i)$	3	4	6	1	8	2	7	5

Starting with 1, and successively applying the function π , we have

$$\pi(1) = 3, \quad \pi(3) = 6, \quad \pi(6) = 2, \quad \pi(2) = 4, \quad \pi(4) = 1.$$

So applying π 5 times has brought 1 back to itself. We write the effect of π in these 5 elements as a **cycle** (13624). This means that π moves each element listed in the cycle to the next one (for example $\pi(3) = 6$) except for the last element, which is moved to the first one ($\pi(4) = 1$). This cycle does not completely determine π , since it does not tell us the effect of π on the elements 5, 7, 8 not listed in the cycle. However we can start again with 5. We find

$$\pi(5) = 8, \quad \pi(8) = 5,$$

so we obtain another cycle (58) of length 2. This just leaves 7, and as $\pi(7) = 7$, this must form a cycle (7) of length 1. We can therefore write

$$\pi = (13624)(58)(7).$$

We usually omit cycles of length 1, so we would write

$$\pi = (13624)(58).$$

Now let π_1 and π_2 be the permutations given by the two cycles separately:

$$\pi_1 = (13624), \quad \pi_2 = (58).$$

(So π_1 does not move 5, 7, 8, and π_2 does not move 1, 2, 3, 4, 6, 7).

These are the permutations shown in the following table:

i	1	2	3	4	5	6	7	8
$\pi_1(i)$	3	4	6	1	5	2	7	8
$\pi_2(i)$	1	2	3	4	8	6	7	5

Since the elements moved by π_1 are not moved by π_2 (and vice versa) we can compose these two permutations either way round to get the original permutation π .

$$\pi_1 \circ \pi_2 = \pi = \pi_2 \circ \pi_1.$$

(Warning: for permutations $\sigma, \tau \in S_n$, it is not usually the case that $\sigma \circ \tau = \tau \circ \sigma$.) So in writing

$$\pi = (13624)(58) \text{ or } \pi = (13624)(58)(7)$$

we have expressed π as a product of disjoint cycles. **Disjoint** means that the cycles have no element in common. We could have chosen different starting points for the cycles, so we could also write

$$\pi = (62413)(85).$$

For convenience, we usually start each cycle with its smallest element.

In the same way, given any permutation $\pi \in S_n$, for any $n \in \mathbb{N}$, we can write π as a product of disjoint cycles. This will always work since, as we repeatedly apply π to find the elements in a given cycle, we will eventually reach a number a already encountered (because there are only finitely many numbers $1, \dots, n$ in the domain of π) and this must be the first number in the current cycle since otherwise π could not be injective.

■ **Example 2.38** There are $3! = 6$ permutations in S_3 . In cycle notation (omitting cycles of length 1) these are

$$\text{id}, \quad (12), \quad (13), \quad (23), \quad (123), \quad (132).$$

We write id for the identity permutation $(1)(2)(3)$ since it contains only cycles of length 1.

2.4.2 Composing permutations in cycle notation

We can compose permutations $\sigma, \tau \in S_n$ to form a new permutation $\sigma \circ \tau$. We usually omit the symbol $\circ$ and just write $\sigma\tau$. We will see how to compose permutations using cycle notation.

Let $\sigma = (234)(56)$ and $\tau = (14)(2365)$. We want to write the permutation $\sigma\tau$ as a product of disjoint cycles. The answer is not just

$$\sigma\tau = (234)(56)(14)(2365)$$

(although this equation is true) since these four cycles are not disjoint: for example the first and third cycles both contain 4. Recall that $\sigma\tau$ means $\sigma \circ \tau$, and $\sigma \circ \tau$ is the function taking each $i \in \{1, \dots, n\}$ to $\sigma(\tau(i))$: we apply the right-hand function τ to i first, giving $\tau(i)$, and then apply σ to this. So to see the effect of $\sigma\tau$ we need to start with the rightmost cycle and work through the cycles from right to left. So to calculate $\sigma\tau(1)$, we observe that the rightmost cycle (2365) leaves 1 as 1, the next cycle (14) moves 1 to 4, the next cycle (56) leaves 4 as 4, and the leftmost cycle (234) moves 4 to 2. So $\sigma\tau(1) = 2$. In a similar way, we find $\sigma\tau(2) = 4$, $\sigma\tau(4) = 1$. This gives the cycle (124) . Continuing with the elements not yet accounted for, we find $\sigma\tau(3) = 5$, $\sigma\tau(5) = (3)$ and $\sigma\tau(6) = 6$. Thus

$$\sigma\tau = (124)(35)(6) = (124)(35).$$

We have now written $\sigma\tau$ as a product of disjoint cycles. If we compose the same permutations the other way round we get

$$\tau\sigma = (14)(2365)(234)(56) = (143)(26).$$

Remark 2.39

This process can be a bit confusing at first, since we take the cycles in order from right to left, but *within* each cycle we work from left to right. (So (2365) moves 2 to 3 and moves 3 to 6, etc.) With a bit of practice, you should be able to do calculations like this quickly and reliably.

Let us check this method does indeed compose permutations correctly by doing the same calculation via tables:

i	1	2	3	4	5	6
$\sigma(i)$	1	3	4	2	6	5
$\tau(i)$	4	3	6	1	2	5
$\sigma\tau(i)$	2	4	5	1	3	6
$\tau\sigma(i)$	4	6	1	3	5	2

To obtain the last two rows of this table, we calculate (using the second and third rows) that $\sigma(\tau(1)) = \sigma(4) = 2$, etc, and $\tau(\sigma(1)) = \tau(1) = 4$, etc.

2.4.3 The inverse of a permutation

Since a permutation is a bijective function, it has an inverse function. This is also a permutation. To find the inverse of permutation written in cycle notation, simply reverse the elements in each cycle (and rewrite to start with the smallest in each cycle). So if

$$\sigma = (234)(56), \quad \tau = (14)(2365)$$

$$\sigma^{-1} = (432)(65) = (243)(56),$$

$$\tau^{-1} = (41)(5632) = (14)(2563).$$

We check

$$\sigma\sigma^{-1} = (234)(56)(243)(56) = (1)(2)(3)(4)(5)(6)(7) = \text{id},$$

and similarly, $\sigma^{-1}\sigma = \text{id}$ and $\tau\tau^{-1} = \text{id} = \tau^{-1}\tau$.

A cycle of length 2 is its own inverse.

2.4.4 The order of a permutation

We can calculate powers of a permutation by repeated multiplication (composition). So if $\sigma = (1\ 2\ 3)$, then

$$\sigma^2 = \sigma\sigma = (1\ 2\ 3)(1\ 2\ 3) = (1\ 3\ 2),$$

$$\sigma^3 = \sigma^2\sigma = (1\ 3\ 2)(1\ 2\ 3) = \text{id},$$

(alternatively, we might notice that $\sigma^2 = \sigma^{-1}$, so that $\sigma^3 = \sigma^{-1}\sigma = \text{id}$). Then $\sigma^4 = \sigma$, $\sigma^5 = \sigma^2$, etc.

For $\pi = (1\ 2\ 4)(5\ 7)(6\ 8)$,

$$\pi^2 = (1\ 2\ 4)(5\ 7)(6\ 8)(1\ 2\ 4)(5\ 7)(6\ 8) = (1\ 4\ 2)$$

$$\pi^3 = (1\ 4\ 2)(1\ 2\ 4)(5\ 7)(6\ 8) = (5\ 7)(6\ 8)$$

$$\pi^4 = (1\ 2\ 4)$$

$$\pi^5 = (1\ 4\ 2)(5\ 7)(6\ 8)$$

$$\pi^6 = \text{id}.$$

Then $\pi^7 = \pi$, $\pi^8 = \pi^2$, etc.

Proposition 2.40 For any permutation $\pi \in S_n$, there is some $m > 0$ such that $\pi^m = \text{id}$ and the permutations $\text{id}, \pi, \pi^2, \dots, \pi^{m-1}$ are all different.

Proof. The sequence of permutations

$$\text{id} = \pi^0, \pi, \pi^2, \pi^3, \dots$$

must eventually contain a repeat, since there are only finitely many (in fact, $n!$) permutations in S_n . If the first repeat occurs after m steps, so $\pi^m = \pi^k$ for some k with $0 \leq k \leq m-1$, then we must have $k = 0$, since otherwise (composing both sides with π^{-1} we would have had a repeat $\pi^{m-1} = \pi^{k-1}$ one step earlier. So $\pi^m = \text{id}$ and there are no repeats amongst $\text{id}, \pi, \pi^2, \dots, \pi^{m-1}$. ■

Clearly m is the *smallest* positive integer such that $\pi^m = \text{id}$. We give it a name:

Definition 2.41 For a permutation π in S_n , the **order** of π is the smallest $m > 0$ such that $\pi^m = \text{id}$.

So $\sigma = (123)$ has order 3 and $\pi = (124)(57)(68)$.

Our next aim is to see how to find the order of a permutation written as a product of disjoint cycles.

First, a preliminary observation. Suppose that $\pi = \sigma_1\sigma_2$, where σ_1, σ_2 are disjoint cycles. Since the order of multiplication of disjoint cycles does not matter, we have

$$\pi^2 = \sigma_1\sigma_2\sigma_1\sigma_2 = \sigma_1\sigma_1\sigma_2\sigma_2 = \sigma_1^2\sigma_2^2,$$

$$\pi^3 = \pi^2\pi = \sigma_1^2\sigma_2^2\sigma_1\sigma_2 = \sigma_1^3\sigma_2^3,$$

and, in general $\pi^k = \sigma_1^k\sigma_2^k$. The same works if π contains more than two cycles: if $\pi = \sigma_1 \dots \sigma_t$, where $\sigma_1, \dots, \sigma_t$ are disjoint cycles, then $\pi^k = \sigma_1^k \dots \sigma_t^k$.

Lemma 2.42 If $\pi = \sigma_1 \dots \sigma_t$, where $\sigma_1, \dots, \sigma_t$ are disjoint cycles of lengths $m_1, \dots, m_t$ respectively, then the order of π is the least com-

mon multiple of these cycle lengths:

$$\text{order of } \pi = \text{lcm}(m_1, \dots, m_t).$$

Proof. For any $k \geq 0$, we have

$$\pi^k = \sigma_1^k \dots \sigma_t^k.$$

Let $j \in \{1, \dots, n\}$.

If j does not occur in any of the cycles $\sigma_1, \dots, \sigma_t$ then $\pi^k(j) = j$.

If j occurs in the cycle σ_u , say, (and hence in no other σ_v , since these cycles are disjoint), then $\pi^k(j) = \sigma_u^k(j)$, and this equals j precisely when j is a multiple of the cycle length m_u (which is the also the order of σ_u).

Thus $\pi^k(j) = j$ for *all* j if and only if j is a multiple of m_u for *each* u . The smallest such $k > 0$ is the least common multiple of $m_1, \dots, m_t$. Hence the order of π is $\text{lcm}(m_1, \dots, m_t)$. ■

■ **Example 2.43** $\pi = (1473)(26598)(1011)$, then the order of π is $\text{lcm}(4, 5, 2) = 20$.

■ **Example 2.44** $\pi = (1234)(5678910)$, then the order of π is $\text{lcm}(4, 6) = 12$.

2.4.5 The sign of a permutation

The simplest non-identity permutations in S_n are those consisting of one length-two cycle (and $n - 2$ length-one cycles). These are called *transpositions* and can be written as (ij) where $1 \leq i < j \leq n$. The transpositions in S_5 are

$$(12), (13), (14), (15), (23), \\ (24), (25), (34), (35), (45).$$

A transposition (ij) swaps i and j and leaves everything else the same.

Every permutation can be written as a product of transpositions.

The special case of a cycle can be done as

$$(a_1 a_2 a_3 \cdots a_k) = (a_1 a_2)(a_2 a_3)(a_3 a_4) \cdots (a_{k-1} a_k).$$

In general, we can do this for each cycle. As an example, inside S_8 ,

$$\begin{aligned} (13874)(265) &= (13)(38)(87)(74)(26)(65) \\ &= (13)(38)(78)(47)(26)(56). \end{aligned}$$

For any given permutation, there are lots of ways to express it as a product of transpositions. As an example,

$$\begin{aligned} (1234) &= (12)(23)(34) = (12)(23)(24)(24)(34) \\ &= (12)(24)(34)(34)(23) = (12)(24)(23) \\ &= (12)(34)(23)(23)(24) = (34)(12)(24) \end{aligned}$$

etc. In this example, we have expressed the permutation as a product of three transpositions and of five transpositions. It's easy to throw in pairs of transpositions, so we could have seven transpositions, nine transpositions etc. But can we use an **even** number of transpositions?

It turns out we cannot.

Theorem 2.45 A permutation cannot be expressed as both a product of an even number of transpositions and of an odd number of transpositions.

A proof is outlined in a tutorial sheet question.

Permutations come in two distinct flavours: **even** permutations which are products of an even number of transpositions, and **odd**

permutations which are products of an odd number of transpositions. So $(13874)(265) = (13)(38)(87)(74)(26)(65)$ is an even permutation and $(1234) = (12)(23)(34)$ is an odd permutation.

The product of two even or two odd permutations is even, and the product of an odd permutation and an even permutation is odd.

A cycle of length k can be written as a product of $k - 1$ transpositions. A cycle of even length is an odd permutation, and a cycle of odd length is an even permutation.

An equivalent notion to evenness/oddness is the *sign* of a permutation. The sign of $\sigma \in S_n$ is denoted $\text{sgn}(\sigma)$ and is defined by

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Equivalently, $\text{sgn}(\sigma) = (-1)^k$ whenever σ is a product of k transpositions. Then the product rule for evenness/oddness simply becomes

$$\text{sgn}(\sigma\rho) = \text{sgn}(\sigma) \text{sgn}(\rho).$$

3 ELEMENTARY NUMBER THEORY

In this section, we will look at some ideas which relate specifically to the *natural numbers*

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

and the integers *integers*

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

Our goal is to give a proof of the fundamental fact that every natural number has factorisation into prime number, and that this factorisation is (in a suitable sense) unique.

We will take for granted the usual algebraic properties of $\mathbb{N}$ and $\mathbb{Z}$. For example, every $m \in \mathbb{Z}$ has an additive inverse $-m$, so for $m, n \in \mathbb{Z}$ we always have

$$n - m = n + (-m) \in \mathbb{Z}.$$

3.1 Proof by Induction

Proof by Induction is a method of proof which is specific to the natural numbers. Suppose we have some statement which wish to prove is true for every natural number n . (The statement might be something we suspect is true because we think we have “spotted a pattern” in several examples, or may be given in a question starting “Prove by induction that ...”.) If we know that

- the statement is true for $n = 1$;
- whenever the statement is true for $n = k$ (for some unspecified natural number k) it is also true for $n = k + 1$;

then the statement is true for all natural numbers n .

A Proof by Induction therefore has the pattern:

- **Check** the statement is true for $n = 1$. (Induction base)
- **Assume** that the statement is true for $n = k$. (Induction hypothesis)
- **Deduce** from the Induction Hypothesis that the statement is also true for $n = k + 1$. (Induction Step)
- **Conclude** that the statement is true for all natural numbers n .

We can state this in a more formal way as the **Principle of Mathematical Induction**: If P is a predicate on $\mathbb{N}$ (so, for each $n \in \mathbb{N}$, the proposition $P(n)$ is either true or false), and we know

(i) $P(1)$ is true;

(ii) $P(k) \Rightarrow P(k + 1)$ for every $k \in \mathbb{N}$;

we can deduce that $P(n)$ is true for all n .

An equivalent statement is the **Well-Ordering Principle**: Every non-empty subset of the natural numbers contains a least element.

■ **Example 3.1** *The sum of the first n odd numbers is n^2 .*

This can be written as

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

or, using summation notation,

$$\sum_{i=1}^n (2i - 1) = n^2. \quad (3.1)$$

When $n = 1$, the left-hand side of (3.1) is just 1 and the right-hand side is $1^2 = 1$, so the equation (3.1) is true for $n = 1$. [Induction Base]

Now suppose that (3.1) is true for $n = k$:

$$\sum_{i=1}^k (2i - 1) = k^2. \quad (3.2)$$

[Induction Hypothesis]

We have to deduce from (3.2) that (3.1) is still true for $n = k + 1$.

(This is the Induction Step.) But for $n = k + 1$,

$$\begin{aligned}
 \text{LHS of (3.1)} &= \sum_{i=1}^{k+1} (2i - 1) \\
 &= \left(\sum_{i=1}^k (2i - 1) \right) + (2(k + 1) - 1) \\
 &= k^2 + (2k + 1) \text{ using (3.2)} \\
 &= (k + 1)^2 \\
 &= \text{RHS of (3.1)}
 \end{aligned}$$

so that (3.1) is indeed true for $n = k + 1$. (Notice that we had to split up the sum of $k + 1$ terms by separating the last term, so that we could make use of (3.2).) So we have completed the induction step.

Hence we can conclude that

$$\sum_{i=1}^n (2i - 1) = n^2$$

for all natural numbers n .

■ **Example 3.2** Prove by induction that

$$\sum_{i=1}^n i^2 = \frac{1}{6}n(n + 1)(2n + 1). \quad (3.3)$$

When $n = 1$, the left-hand side of (3.3) is just $1^2 = 1$ and the right-hand side is

$$\frac{1}{6} \times 1 \times 2 \times 3 = 1,$$

so (3.3) is true for $n = 1$.

Now suppose that (3.3) is true for $n = k$:

$$\sum_{i=1}^k i^2 = \frac{1}{6}k(k + 1)(2k + 1). \quad (3.4)$$

Then, for $n = k + 1$,

$$\begin{aligned}
 \text{LHS of (3.3)} &= \sum_{i=1}^{k+1} i^2 \\
 &= \left(\sum_{i=1}^k i^2 \right) + (k+1)^2 \\
 &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\
 &= \frac{1}{6} \left[k(k+1)(2k+1) + 6(k+1)^2 \right] \\
 &= \frac{1}{6}(k+1) \left[2k^2 + k + 6(k+1) \right] \\
 &= \frac{1}{6}(k+1) (2k^2 + 7k + 6) \\
 &= \frac{1}{6}(k+1)(k+2)(2k+3) \\
 &= \text{RHS of (3.3)}
 \end{aligned}$$

So (3.3) is again true for $n = k + 1$. Hence

$$\sum_{i=1}^n i^2 = \frac{1}{6}n(n+1)(2n+1) \text{ for all natural numbers } n.$$

Proof by Induction can be used for many things apart from summation formulae.

■ **Example 3.3** *Prove by induction that $9^n - 8n - 1$ is a multiple of 64 for every natural number n .*

For $n = 1$ we have $9^n - 8n - 1 = 9^1 - 8 - 1 = 0 = 0 \times 64$, so the statement is true for $n = 1$.

Suppose the statement is true for $n = k$, so

$$9^k - 8k - 1 = 64a \text{ for some } a \in \mathbb{Z} \text{ (depending on } k\text{).}$$

We can rewrite this as

$$9^k = 8k + 1 + 64a.$$

Now consider the statement for $n = k + 1$:

$$\begin{aligned}
 9^{k+1} - 8(k+1) - 1 &= 9 \times 9^k - 8k - 9 \\
 &= 9(8k + 1 + 64a) - 8k - 9 \\
 &= 64k + 9 \times 64a \\
 &= 64(k + 9a).
 \end{aligned}$$

Since $k + 9a \in \mathbb{Z}$, this means that $9^{k+1} - 8(k+1) - 1$ is a multiple of 64. Thus the statement is also true for $n = k + 1$.

Hence, by induction, $9^n - 8n - 1$ is a multiple of 64 for all $n \geq 1$.

Sometimes we start the induction at a point other than 1:

■ **Example 3.4** Show that, for $n \geq 3$, the sum of the interior angles of an n -sided polygon is $180(n - 2)^\circ$.

(Note that the polygon does *not* have to be regular; the sides and angles need not all be equal.)

Here, the statement does not even make sense for $n = 1$ or $n = 2$: a polygon must have at least 3 sides.

When $n = 3$, the statement says that the sum of the angles in a triangle (3-sided polygon) is 180° , which is well-known to be true (and not proved here!)

Now suppose the statement is true for $n = k$ (where k is some natural number with $k \geq 3$), and consider any $k + 1$ -sided polygon. Label its vertices (in order) as $P_1, P_2, \dots, P_{k+1}$, choosing to start the numbering so that the line P_1P_k is *inside* the polygon. (This line would lie outside the polygon if the interior angle at P_{k+1} is greater

than 180° .) Then

$$\begin{aligned}
 \text{Angle sum of } P_1 \dots p_{k+1} &= \text{Angle sum of } P_1 \dots P_k + \\
 &\quad \text{Angle sum of triangle } P_k P_{k+1} P_1 \\
 &= 180(k-2)^\circ + 180^\circ \\
 &= 180((k+1)-2)^\circ,
 \end{aligned}$$

so the statement is true for $n = k + 1$. (You might find it helpful to draw a diagram.)

Hence, by induction, it is true for all $n \geq 3$. ■

[This is equivalent to an ordinary induction proof, starting at $n = 1$, but for the statement that the sum of the interior angles of an $(n + 2)$ -sided polygon is $180n^\circ$.]

A variant of Induction is **Strong Induction**, where we show that the statement is true for $n = k + 1$ under the assumption that it is true for *all* $n \leq k$ (not just for $n = k$).

■ **Example 3.5** A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_1 = 1; \quad a_2 = 0;$$

$$a_n = -2a_{n-1} - a_{n-2} \text{ for } n \geq 3.$$

(So $a_3 = -2a_2 - a_1 = -1$, $a_4 = -2a_3 - a_2 = 2$, etc.)

Prove by (strong) induction that $a_n = (-1)^n(n-2)$ for all $n \geq 1$.

Here, since the definition of a_n (for all but the first two values of n) is given by a recurrence relation involving the two previous terms, our induction step will need to assume the formula for both $n = k$ and $n = k - 1$ in order to show it for $n = k + 1$. Since the first two values of n are treated separately in the definition of the sequence, we will need to treat both of them separately in the proof, so the base

of our induction will involve both $n = 1$ and $n = 2$.

When $n = 1$, the formula gives $a_1 = (-1)^1(1 - 2) = -(-1) = 1$, which is the correct value. When $n = 2$, it gives $a_2 = (-1)^2(2 - 2) = 0$, which is again correct,

Now suppose that $k \geq 2$ and the formula is correct for all $n \leq k$. (We have just checked that this is the case for $k = 2$.) We will check the formula is also correct for $n = k + 1$.

We know in particular that the formula is correct for $n = k$ and $n = k - 1$, so that

$$a_k = (-1)^k(k - 2), \quad a_{k-1} = (-1)^{k-1}(k - 3).$$

Now $k + 1 \geq 3$, so from the recurrence relation we have

$$\begin{aligned} a_{k+1} &= -2a_k - a_{k-1} \\ &= -2(-1)^k(k - 2) - (-1)^{k-1}(k - 3) \\ &= 2(-1)^{k+1}(k - 2) - (-1)^{k-1}(k - 3) \\ &= 2(-1)^{k+1}(k - 2) - (-1)^{k+1}(k - 3) \\ &= (-1)^{k+1}[2(k - 2) - (k - 3)] \\ &= (-1)^{k+1}[k - 1] \\ &= (-1)^{k+1}[(k + 1) - 2], \end{aligned}$$

so the formula is also true for $n = k + 1$. Hence, by (strong) induction, it is true for natural numbers:

$$a_n = (-1)^n(n - 2) \text{ for all } n \geq 1.$$

Why does strong induction work? Again, we are in effect applying the standard version of induction to a modified statement. In this

case, the statement which we are proving for each n is that

$$a_m = (-1)^m(m-2) \text{ for all } m \text{ with } 1 \leq m \leq n.$$

For $n = 1$, we checked this by verifying the formula for $n = 1$. If we assume the statement for $n = k \geq 1$, then either $k = 1$ and we must check a_2 separately (since the recurrence relation does not apply), or $k > 1$ and a_{k+1} is given by the recurrence relation. Our induction hypothesis then tells us that the formula is correct for a_k and a_{k-1} , so we can check it also holds for a_{k+1} .

Finally, here is an example where the statement to be proved is not given in advance.

■ **Example 3.6** Which is bigger, 2^n or n^2 ?

To investigate this, make a table for the first few values of n .

n	1	2	3	4	5	6	7
2^n	2	4	8	16	32	64	128
n^2	1	4	9	16	25	36	49

For small n , the answer varies: $2^n > n^2$ for $n = 1$, but $n^2 > 2^n$ for $n = 3$, while $2^n = n^2$ for $n = 2$ and $n = 4$. However, $2^n > n^2$ for $n = 5, 6, 7$, and the table suggests that 2^n continues to grow faster than n^2 . So the table leads us to *guess* that $2^n > n^2$ for *all* $n \geq 5$. We can prove this guess really is correct using induction.

The statement we want to prove is: $2^n > n^2$ for all natural numbers $n \geq 5$.

We have already seen the statement is true when $n = 5$: $2^5 = 32 > 25 = 5^2$.

Now suppose that we know $2^k > k^2$ for some $k \geq 5$. Then

$$\begin{aligned}
 2^{k+1} &= 2 \times 2^k \\
 &> 2 \times k^2 \text{ by our induction hypothesis} \\
 &= k^2 + k^2 \\
 &\geq k^2 + 5k \text{ since } k \geq 5 \\
 &= k^2 + 2k + 3k \\
 &> k^2 + 2k + 1 \\
 &= (k+1)^2.
 \end{aligned}$$

This shows that the statement is again true for $n = k + 1$. So, by induction, we have proved that $2^n > n^2$.

In this example, we have the added complication that we are dealing with inequalities rather than equations. We have to spot a way to get from 2^{k+1} to $(k+1)^2$ by estimating parts of the expressions involved, making sure the inequalities all go in the right direction.

3.2 Prime Numbers and Divisibility

We want to understand the structure of the natural numbers, in particular the concept of *prime number*. Our goal in this section is to give a proof of the important fact that every natural number has a *unique* prime factorisation.

Definition 3.7 A *prime number* is a natural number p which is greater than 1, and which is only divisible by the natural numbers 1 and p .

Note that 1 is **not** a prime number.

To make sense of this, we need to have a firm understanding of the notion of divisibility.

Definition 3.8 Let $m, n \in \mathbb{Z}$. Then n is *divisible by m* if there exists $a \in \mathbb{Z}$ such that $n = ma$.

For example,

- 91 is divisible by 7, as $91 = 7 \times 13$,
- 45 is divisible by -5 , as $45 = (-5) \times (-9)$,
- 87 is not divisible by 11, because there is no integer a with $87 = 11a$ (as $7 < 87/11 < 8$).
- Every number is divisible by 1.
- A number is divisible by 2 if and only if it is even.
- A number is divisible by 5 if and only if it ends in 0 or 5, and is divisible by 10 if and only if it ends in 0.

If n is divisible by m we say that n is a *multiple of m* or that m is a *divisor of n* , or m is a *factor of n* , or just that m *divides n* .

We write $m \mid n$ when m divides n (i.e. n is divisible by m ; note the change of order). The symbol $\mid$ means "divides" or "goes into".

We also write $m \nmid n$ when n is not divisible by m .

For example $6 \mid 30$ but $7 \nmid 31$.

Remark 3.9

Don't confuse the symbols $\mid$ and $/$.

Provided that $m \neq 0$, the statement $m \mid n$ means that the number n/m (or $\frac{n}{m}$) belongs to $\mathbb{Z}$.

A prime number p can't have any divisors other than 1 and p . So, an integer $n > 1$ is prime if none of the numbers d with $1 < d < n$ are divisors of n . For example, 5 is prime since none of 2, 3 and 4 are divisors of 5, but 6 is not prime as, for instance, $3 \mid 6$.

Here is the first basic property of primes.

Theorem 3.10 Let n be a natural number with $n > 1$. Then n has a prime factor p .

Proof. We use Strong Induction, starting at $n = 2$. So we assume

$$\forall m < n : \text{if } m > 1 \text{ then } m \text{ has a prime factor,}$$

and we need to show that n has a prime factor. There are two cases.

- If n is prime (which includes the base case $n = 2$) then we can take $p = n$.

- If n is not prime (but $n > 1$), then n has a factor m with $1 < m < n$.

By the strong induction hypothesis, m has a prime factor p , which is also a prime factor of n .

■

The sequence of primes starts 2, 3, 5, 7, 11, 13, ...

How many primes are there?

Theorem 3.11 — Euclid. There are infinitely many prime numbers.

Proof. Suppose for a contradiction that there are only finitely many primes: $p_1, \dots, p_k$. Define $N = p_1 p_2 \cdots p_k + 1$. Each p_j divides $p_1 p_2 \cdots p_k$, so $p_j \nmid N$. But N has a prime factor p , and $p \neq p_j$ for all j . Contradiction! ■

The Sieve of Eratosthenes is a basic method for generating all primes up to a given bound N .

- Write down the numbers from 2 to N inclusive.
- The first number is 2. Highlight it and delete all other multiples of 2.
- The first unhighlighted number is now 3. Highlight it and delete all other multiples of 3.

- Keep going! At each stage, highlight the first remaining unhighlighted number, and delete all its other multiples.

Eventually all the numbers left are highlighted. They are the primes between 2 and N inclusive.

■ **Example 3.12** We find all the primes up to 120. First, we write down the numbers from 2 to 120:

	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80	81	82	83	84
85	86	87	88	89	90	91	92	93	94	95	96
97	98	99	100	101	102	103	104	105	106	107	108
109	110	111	112	113	114	115	116	117	118	119	120

We then highlight 2 and remove all other multiples of 2:

	2	3	5	7	9	11
13		15	17	19	21	23
25		27	29	31	33	35
37		39	41	43	45	47
49		51	53	55	57	59
61		63	65	67	69	71
73		75	77	79	81	83
85		87	89	91	93	95
97		99	101	103	105	107
109		111	113	115	117	119

The first unhighlighted number remaining is 3, so we highlight 3 and remove all other multiples of 3:

	2	3	5	7	11
13			17	19	23
25			29	31	35
37			41	43	47
49			53	55	59
61			65	67	71
73			77	79	83
85			89	91	95
97			101	103	107
109			113	115	119

Next we highlight 5, and remove all other multiples of 5. We then do the same for 7.

	2	3	5	7	11
13			17	19	23
			29	31	
37			41	43	47
49			53		59
61				67	71
73			77	79	83
			89	91	
97			101	103	107
109			113		119

	2	3	5	7	11
13			17	19	23
			29	31	
37			41	43	47
			53		59
61				67	71
73				79	83
			89		
97		101	103	107	
109		113			

The next number to highlight is 11, but there are no other multiples of 11 to remove. In fact, all the remaining numbers are prime, since if $2 \leq n \leq 120$ and n is not prime, then n must have at least one prime factor < 11 because $11^2 > n$. We can therefore highlight all the remaining numbers.

	2	3	5	7	11
13			17	19	23
			29	31	
37			41	43	47
			53		59
61				67	71
73				79	83
			89		
97		101	103	107	
109		113			

We have now found the first thirty prime numbers.

I claim that every number bigger than 1 is a prime or a product of primes. For example

- $138 = 2 \times 3 \times 23$;

- $328 = 2 \times 2 \times 2 \times 41 = 2^3 \times 41$;
- $169 = 13^2$;
- 173 is prime.

As we see with 328 and 169, prime factors may occur more than once. We can bundle together equal primes into prime powers.

We now give a proof of the claim.

Theorem 3.13 If n is a natural number, with $n > 1$, then either n is prime, or is a product of a (finite) sequence of primes.

Proof. We employ Strong Induction. Assume that $n > 1$, and that all numbers m with $1 < m < n$ are primes or products of primes. If n is a prime the conclusion certainly holds. Otherwise, n has a prime factor p (by Theorem 3.10) and $1 < p < n$. Then $m = n/p$ is an integer, and $1 < m < n$. By the inductive hypothesis, m is prime, or a product of primes, and it follows that $n = pm$ is too. ■

Experiments with small numbers suggest that prime factorization is “unique”. We can write prime factorizations in several ways, for instance

$$60 = 2 \times 2 \times 3 \times 5 = 5 \times 2 \times 3 \times 2 = 3 \times 5 \times 2 \times 2$$

etc. But all these are just re-arrangements of each other. The big theorem is that all prime factorizations are the same up to order. Alternatively, in two prime factorizations for the same number, each prime occurring in one side, also occurs on the other side, and indeed occurs the same number of times. Later we prove:

Theorem 3.14 — Fundamental Theorem of Arithmetic. Let n have the prime factorizations

$$n = p_1 \cdots p_r = q_1 \cdots q_s.$$

Then every prime occurs equally often in both factorizations (and so $r = s$).

3.3 Modular arithmetic

To work up to proving unique factorization we need to use *modular arithmetic*.

Definition 3.15 For an integer $n \geq 1$, we say that integers a and b are **congruent modulo n** , whenever $(a - b)/n$ is an integer.

We write $a \equiv b \pmod{n}$ when a is congruent to b modulo n .

For example $17 \equiv 9 \pmod{4}$ as $(17 - 9)/4 = 2$ is an integer. But $19 \not\equiv 7 \pmod{5}$ as $(19 - 7)/5 = 12/5$ is not an integer.

Congruence modulo n behaves like equality. (It is an equivalence relation.)

Theorem 3.16 • $a \equiv a \pmod{n}$ for all a (so congruence modulo n is *reflexive*),

- if $a \equiv b \pmod{n}$ then $b \equiv a \pmod{n}$ (so congruence modulo n is *symmetric*),
- if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$ (so congruence modulo n is *transitive*).

Proof. These are routine deductions from the definition. Given $a \in \mathbb{Z}$, $(a - a)/n = 0$ is an integer, so $a \equiv a \pmod{n}$.

If $a \equiv b \pmod{n}$ then $r = (a - b)/n \in \mathbb{Z}$. Then $-r = (b - a)/n \in \mathbb{Z}$ too, so $b \equiv a \pmod{n}$.

If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $r = (a - b)/n \in \mathbb{Z}$

and $s = (b - c)/n \in \mathbb{Z}$. Then $r + s = (a - c)/n \in \mathbb{Z}$ too, so $a \equiv c \pmod{n}$. ■

Remark 3.17

For a fixed n , congruence modulo n is an *equivalence relation* on $\mathbb{Z}$.

A *relation* on a set S is a property which is either true or false for any pair of elements of S . (More formally, it is a predicate on the Cartesian product $S \times S$, and can be identified with the subset of $S \times S$ on which this property holds.) Some examples of relations on $\mathbb{R}$ are $=$, $<$, “has the same square as”, $\neq$, since, given $x, y \in \mathbb{R}$, each of the statements $x = y$, $x < y$, $x^2 = y^2$, $x \neq y$ is either true or false.

A relation $\sim$ on S is an *equivalence relation* if it has the three properties of symmetry, reflexivity and transitivity in Theorem 3.16.

Up to a point, congruences preserve arithmetic operations:

Theorem 3.18 Suppose that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then

- $a + c \equiv b + d \pmod{n}$,
- $a - c \equiv b - d \pmod{n}$ and
- $ac \equiv bd \pmod{n}$.

Proof. Suppose that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Then $r = (a - b)/n \in \mathbb{Z}$ and $s = (c - d)/n \in \mathbb{Z}$. This means that $a = b + rn$ and $c = d + sn$. Therefore

$$\frac{(a + c) - (b + d)}{n} = \frac{b + rn + d + sn - b - d}{n} = r + s \in \mathbb{Z}.$$

Therefore $a + c \equiv b + d \pmod{n}$. A very similar (omitted) argument

gives $a - c \equiv b - d \pmod{n}$. Finally,

$$\begin{aligned} \frac{ac - bd}{n} &= \frac{(b + rn)(d + sn) - bd}{n} \\ &= \frac{bsn + drn + rsn^2}{n} = bs + dr + rsn \in \mathbb{Z}. \end{aligned}$$

Therefore $ac \equiv bd \pmod{n}$. ■

Remark 3.19

One has to be careful with division: it does not always respect congruences. We may have $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ but $a/c \not\equiv b/d \pmod{n}$, even when both a/c and b/d are integers.

For example

$$36 \equiv 16 \pmod{10} \quad \text{and} \quad 4 \equiv 4 \pmod{10}$$

but

$$\frac{36}{4} \not\equiv \frac{16}{4} \pmod{10}$$

as $9 \not\equiv 4 \pmod{10}$.

Given $n \geq 1$, each number $a \in \mathbb{Z}$ is congruent modulo n to precisely one number in the set $\{0, 1, \dots, n-1\}$.

Theorem 3.20 Let $n \geq 1$ and $a \in \mathbb{Z}$. There are unique $q \in \mathbb{Z}$ and $r \in \{0, 1, \dots, n-1\}$ with $a = qn + r$. (We call q the *quotient* and r the *remainder* obtained by dividing a by n .)

Proof. To show the existence of q and r , we divide a by n , getting a rational number $a/n \in \mathbb{Q}$. We then “round down” a/n to an integer q , so that $q \leq a/n < q+1$. Then $qn \leq a < qn+n$, so if we let $r = a - qn$ then $a \equiv r \pmod{n}$ and $0 \leq r < n$.

We now need to show uniqueness. Suppose then that $a = qn + r = q'n + r'$ with $q, q' \in \mathbb{Z}$ and $r, r' \in \{0, 1, \dots, n-1\}$. Then $r \equiv r' \pmod{n}$. But as $0 \leq r < n$ and $0 \leq r' < n$, then $-n < r - r' < n$

and $-1 < (r - r')/n < 1$. But $(r - r')/n$ is an integer, and the only integer between -1 and 1 is 0 . Thus $(r - r')/n = 0$ and so $r = r'$. It then follows that $q = q'$. ■

■ **Example 3.21** Let $n = 8$, so each number is congruent modulo 8 to exactly one number in $\{0, 1, 2, 3, 4, 5, 6, 7\}$. For instance

$$100 \equiv 4 \pmod{8}, \quad 77 \equiv 5 \pmod{8}, \quad -38 \equiv 2 \pmod{8}.$$

In the first case, $100/8 = 25/2$ which is between 12 and 13 , so $q = 12$ and $100 \equiv 100 - 12 \times 8 = 4 \pmod{8}$ etc.

The set of numbers congruent modulo n to a given number c is the *congruence class of c modulo n* . The congruence class of c is denoted by $[c]_n$ (or sometimes just $[c]$).

For example $[1]_2$ is the set of odd numbers.

A more interesting example:

$$[13]_5 = \{\dots, -22, -17, -12, -7, -2, 3, 8, 13, 18, 23, \dots\}.$$

This also equals $[-12]_5$ and $[123]_5$ etc.

The congruence classes $[a]_n$ and $[b]_n$ are the same if and only if $a \equiv b \pmod{n}$. So for each n , there are precisely n distinct congruence classes modulo n , namely $[0]_n, [1]_n, \dots, [n-1]_n$.

For example, when $n = 4$ we have the following congruence classes:

$$[0]_4 = \{\dots, -12, -8, -4, 0, 4, 8, 12, 16, \dots\},$$

$$[1]_4 = \{\dots, -11, -7, -3, 1, 5, 9, 13, 17, \dots\},$$

$$[2]_4 = \{\dots, -10, -6, -2, 2, 6, 10, 14, 18, \dots\},$$

$$[3]_4 = \{\dots, -9, -5, -1, 3, 7, 11, 15, 19, \dots\}.$$

Therefore $[0]_4 = [4]_4 = [-12]_4$ etc., and $[3]_4 = [-9]_4 = [19]_4$ etc.

We can add, subtract and multiply congruence classes:

$$[a]_n + [b]_n = [a + b]_n, \quad [a]_n - [b]_n = [a - b]_n, \quad [a]_n [b]_n = [ab]_n.$$

This makes sense: if $[a]_n = [a']_n$ and $[b]_n = [b']_n$ then $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$ and so $a + b \equiv a' + b' \pmod{n}$ and so $[a + b]_n = [a' + b']_n$. The same is true for subtraction and multiplication.

For example $[4]_9 + [6]_9 = [10]_9 = [1]_9$, $[4]_9 - [6]_9 = [-2]_9 = [7]_9$ and $[4]_9 [6]_9 = [24]_9 = [6]_9$.

The set

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}$$

is a set where we have well-defined operations of addition, subtraction and multiplication, and is called the *integers modulo n*.

Here are the addition and multiplication tables for $\mathbb{Z}_5$. I write $[a]$ for $[a]_5$.

+	[0]	[1]	[2]	[3]	[4]	×	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]	[0]	[0]	[0]	[0]	[0]	[0]
[1]	[1]	[2]	[3]	[4]	[0]	[1]	[0]	[1]	[2]	[3]	[4]
[2]	[2]	[3]	[4]	[0]	[1]	[2]	[0]	[2]	[4]	[1]	[3]
[3]	[3]	[4]	[0]	[1]	[2]	[3]	[0]	[3]	[1]	[4]	[2]
[4]	[4]	[0]	[1]	[2]	[3]	[4]	[0]	[4]	[3]	[2]	[1]

Here is just the multiplication table for $\mathbb{Z}_6$. I write $[a]$ for $[a]_6$.

$\times$	[0]	[1]	[2]	[3]	[4]	[5]
[0]	[0]	[0]	[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]	[3]	[4]	[5]
[2]	[0]	[2]	[4]	[0]	[2]	[4]
[3]	[0]	[3]	[0]	[3]	[0]	[3]
[4]	[0]	[4]	[2]	[0]	[4]	[2]
[5]	[0]	[5]	[4]	[3]	[2]	[1]

For addition, $[0]_n$ is always an *identity* for $\mathbb{Z}_n$:

$$[a]_n + [0]_n = [a]_n.$$

For multiplication, the identity is $[1]_n$:

$$[a]_n [1]_n = [a]_n.$$

For addition, each $[a]_n$ has an *inverse*, an element which when added to $[a]_n$ gives the (additive) identity. This is $[-a]_n$ (which is the same as $[n - a]_n$):

$$[a]_n + [-a]_n = [0]_n.$$

Inverses under multiplication are more interesting. The multiplicative inverse of $[a]_n$ will be $[b]_n$ provided that $[a]_n [b]_n = [1]_n$ (equivalently $ab \equiv 1 \pmod{n}$), since $[1]_n$ is the multiplicative identity. Clearly $[0]_n$ has no inverse since $[0]_n [b]_n = [0]_n$, but in $\mathbb{Z}_5$ every other element has an inverse: $[1]_5$ and $[4]_5$ are their own inverses, while $[2]_5$ and $[3]_5$ are each other's inverse. In $\mathbb{Z}_5$ every nonzero element has an inverse.

But this is not the case in $\mathbb{Z}_6$. Only $[1]_6$ and $[5]_6$ have inverses there (each is its own inverse).

(I'm now being lazy and writing $[a]$ for $[a]_n$.)

These questions now arise

- When does $[a]$ have an inverse in $\mathbb{Z}_n$?
- Which n have the property that all nonzero elements of $\mathbb{Z}_n$ have inverses?

Let's collect some evidence: if $n = 10$ then no $[a]$ with a even is invertible, since ab will be even for all $b \in \mathbb{Z}$ and $ab \not\equiv 1 \pmod{10}$. Also $[5]$ is not invertible as then $5b \equiv 0 \pmod{5}$ so that $5b \not\equiv 1 \pmod{10}$. But $[1], [3], [7], [9]$ have inverses. We can tabulate these:

element	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
inverse	-	[1]	-	[7]	-	-	-	[3]	-	[9]

Let's do the same for $\mathbb{Z}_{11}$. We could do that by completing the multiplication table, and finding the $[1]$ s therein, but I'll use an *ad hoc* approach.

Of course, $[1]$ is its own inverse. As $12 \equiv 1 \pmod{11}$ then $2 \times 6 = 12 \equiv 1 \pmod{11}$. Therefore $[3][4] = [1]$ and $[2][6] = [1]$. But also $(-3)(-4) \equiv 1 \pmod{11}$, so $[-3][-4] = [1]$, equivalently $[8][7] = [1]$. Similarly, $[9][5] = [1]$ and $[10][10] = [1]$. Collecting these we get:

element	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
inverse	-	[1]	[6]	[4]	[3]	[9]	[2]	[8]	[7]	[5]	[10]

Of our examples so far, in $\mathbb{Z}_5$ and $\mathbb{Z}_{11}$ all nonzero elements have inverses. Also 5 and 11 are primes. Coincidence?

We can partly characterize invertible elements in $\mathbb{Z}_n$.

Theorem 3.22 If some number d divides both a and n , and $d > 1$, then $[a]$ is **not** invertible in $\mathbb{Z}_n$.

Proof. We want to show that there is no integer b with $ab \equiv 1 \pmod{n}$. If there was such a number b then $ab - 1 = cn$ for $c \in \mathbb{Z}$, that is $ab - cn = 1$. But $d \mid a$ and $d \mid n$ so that $d \mid ab$ and $d \mid cn$, and then $d \mid (ab - cn)$, so $d \mid 1$, which is impossible. This contradiction means that $[a][b]$ cannot equal $[1]$. ■

As a consequence, if $n > 1$ isn't prime, then $\mathbb{Z}_n$ has nonzero non-invertible elements: just write $n = am$ where a, m are integers > 1 . Then $[a]$ has no inverse in $\mathbb{Z}_n$.

Can we prove the converse of this theorem? Is $[a]$ invertible in $\mathbb{Z}_n$ whenever a and n have no common divisor d with $d > 1$?

We need to enquire into the nature of common divisors.

3.4 Greatest Common Divisors

Let $m, n \in \mathbb{N}$. Then a *common divisor* of m and n is a positive integer d with $d \mid m$ and $d \mid n$. The *greatest common divisor* (sometimes known as the *highest common factor*) of m and n is the largest number which is a common factor of m and n . We write the greatest common divisor of m and n as $\gcd(m, n)$.

I hope it is apparent that always $\gcd(n, m) = \gcd(m, n)$.

As an example, consider $m = 30$ and $n = 48$.

The divisors of 30 are 1, 2, 3, 5, 6, 10, 15 and 30.

The divisors of 48 are 1, 2, 3, 4, 6, 8, 12, 16, 24 and 48.

The common divisors of 30 and 48 are 1, 2, 3 and 6

The greatest common divisor of 30 and 48 is $\gcd(30, 48) = 6$.

This example demonstrates that if we can find all factors of m and n then we can find $\gcd(m, n)$. Also if we can find the prime factorizations of m and n we can find $\gcd(m, n)$.

In ancient Greece, Euclid gave an algorithm to find $\gcd(m, n)$ without factorizing m and n . Euclid's algorithm is one of the main foundations of algebra and number theory. You may ask, with modern computers why use Euclid's algorithm instead of factorization?

- factorization into primes remains a hard problem even with modern computer technology,
- advanced factorization methods **use** the Euclidean algorithm ex-

tensively.

The key to Euclid's algorithm is the following lemma.

Lemma 3.23 Let $m, n \in \mathbb{N}$. Then for any integer k

$$\gcd(n, m - kn) = \gcd(m, n).$$

Proof. The idea is to show that the common divisors of m and n are precisely the same as the common divisors of n and $m - kn$.

Let d be a common divisor of m and n . Then $m \equiv 0$ and $n \equiv 0 \pmod{d}$. Therefore $m - kn \equiv 0 - k \times 0 = 0 \pmod{d}$. So d is a divisor of $m - kn$ and so a common divisor of n and $m - kn$.

Conversely, let d be a common divisor of n and $m - kn$. Then $n \equiv 0$ and $m - kn \equiv 0 \pmod{d}$. This implies that $m - k \times 0 \equiv 0 \pmod{d}$. Thus $m \equiv 0 \pmod{d}$, and $d \mid m$. So d is a common divisor of m and n . ■

■ **Example 3.24** Let's compute $\gcd(48, 30)$ in a *ad hoc* way using Lemma 3.23. First note that $48 - 1 \times 30 = 18$. Therefore

$$\gcd(48, 30) = \gcd(30, 48 - 1 \times 30) = \gcd(30, 18).$$

Next, $30 - 1 \times 18 = 12$. Therefore

$$\gcd(30, 18) = \gcd(18, 30 - 1 \times 18) = \gcd(18, 12).$$

Again, $18 - 1 \times 12 = 6$. Therefore

$$\gcd(18, 12) = \gcd(12, 18 - 1 \times 12) = \gcd(12, 6).$$

But now we are in the situation where $6 \mid 12$. All divisors of 6 are also divisors of 12. The common divisors of 12 and 6 are just the divisors of 6, so the greatest common divisor of 12 and 6 is 6. We

conclude that

$$\gcd(48, 30) = \gcd(12, 6) = 6.$$

We systematize this method as the *Euclidean algorithm*.

- We start with positive integers m and n .
- Write $m_0 = m$ and $m_1 = n$.
- Repeat the following step for $k = 1, 2, 3, \dots$
 - Let q_k and r_k be the quotient and remainder obtained by dividing m_{k-1} by m_k , and define m_{k+1} as r_k
- Until we arrive at $r_k = 0$, that is $m_k \mid m_{k-1}$.
- Then $m_k = \gcd(m, n)$.

We repeat Example 3.23 using this more systematic method:

■ **Example 3.25** For $m_0 = 48$ and $m_1 = 30$, we get $q_1 = 1$, $r_1 = 18$, so $m_2 = 18$. Then $q_2 = 1$ and $r_2 = 12$, so $m_3 = 12$. Then $q_3 = 1$ and $r_3 = 6$, so $m_4 = 6$. Then $q_4 = 2$ and $r_4 = 0$, so $\gcd(30, 18) = m_4 = 6$.

We can tabulate this as

k	m_k	q_k	r_k	
0	48			
1	30	1	18	$48 - 1 \times 30 = 18$
2	18	1	12	$30 - 1 \times 18 = 12$
3	12	1	6	$18 - 1 \times 12 = 6$
4	6	2	0	$12 - 2 \times 6 = 0$

To be honest, the k column is not very useful, and the r_k column just carries forward into the m_k column; so you could write this as

m_k	q_k	
48		
30	1	$48 - 1 \times 30 = 18$
18	1	$30 - 1 \times 18 = 12$
12	1	$18 - 1 \times 12 = 6$
6	2	$12 - 2 \times 6 = 0$

Here is a more complex example.

■ **Example 3.26** Let $m = 9499$ and $n = 6986$.

m_k	q_k	
9499		
6986	1	$9499 - 6986 = 2513$
2513	2	$6986 - 2 \times 2513 = 1960$
1960	1	$2513 - 1960 = 553$
553	3	$1960 - 3 \times 553 = 301$
301	1	$553 - 301 = 252$
252	1	$301 - 252 = 49$
49	5	$252 - 5 \times 49 = 7$
7	7	$49 - 7 \times 7 = 0$

Therefore $\gcd(9499, 6986) = 7$.

In our examples, we can express every number m_j in terms of the starting terms m and n , as $a_j m + b_j n$ for some integers a_j and b_j . In Example 3.24 ($m = 48, n = 30$) we had

$$18 = 48 - 30 = m - n,$$

$$12 = 30 - 18 = n - (m - n) = -m + 2n,$$

$$6 = 18 - 12 = (m - n) - (-m + 2n) = 2m - 3n.$$

Theorem 3.27 — Bezout's formula. Let $m, n \in \mathbb{N}$. Then there are integers a and b with

$$am + bn = \gcd(m, n)$$

I won't write out a formal proof of this important result, since it is just an observation that if we can write m_{k-1} and m_k in this form, it follows that $m_{k+1} = m_{k-1} - q_k m_k$ also has this form.

Let's see this in action. We use the same numbers as in Example 3.26.

■ **Example 3.28** Let $m = 9499$ and $n = 6896$. Then

$$m_2 = 2513 = 9499 - 6896 = m - n,$$

$$m_3 = 1960 = 6896 - 2 \times 2513 = n - 2(m - n) = -2m + 3n,$$

$$m_4 = 553 = 2513 - 1960 = (m - n) - (-2m + 3n) = 3m - 4n,$$

$$\begin{aligned} m_5 &= 301 = 1960 - 3 \times 553 = (-2m + 3n) - 3(3m - 4n) \\ &= -11m + 15n, \end{aligned}$$

$$\begin{aligned} m_6 &= 252 = 553 - 301 = (3m - 4n) - (-11m + 15n) \\ &= 14m - 19n, \end{aligned}$$

$$\begin{aligned} m_7 &= 49 = 301 - 252 = (-11m + 15n) - (14m - 19n) \\ &= -25m + 34n, \end{aligned}$$

$$\begin{aligned} m_8 &= 7 = 252 - 5 \times 49 = (14m - 19n) - 5(-25m + 34n) \\ &= 139m - 189n. \end{aligned}$$

In this example we get $a = 139$ and $b = -189$.

We can tabulate Example 3.28 as

m_k	q_k	a_k	b_k	
9499		1	0	
6986	1	0	1	
2513	2	1	-1	$2513 = 9499 - 6986$
1960	1	-2	3	$1960 = 6986 - 2 \times 2513$
553	3	3	-4	$553 = 2513 - 1960$
301	1	-11	15	$301 = 1960 - 3 \times 553$
252	1	14	-19	$252 = 553 - 301$
49	5	-25	34	$49 = 301 - 252$
7	7	139	-189	$7 = 252 - 5 \times 49$

This is the *extended Euclidean algorithm*.

How does this work? The first two lines of a and b values are 1,0 and 0,1, since $m_0 = m = 1m + 0n$ and $m_1 = n = 0m + 1n$. In general

$$\begin{aligned}
 m_{k+1} &= m_{k-1} - q_k m_k = (a_{k-1}m + b_{k-1}n) - q_k(a_k m + b_k n) \\
 &= (a_{k-1} - q_k a_k)m + (b_{k-1} - q_k b_k)n
 \end{aligned}$$

so that

$$a_{k+1} = a_{k-1} - q_k a_k \quad \text{and} \quad b_{k+1} = b_{k-1} - q_k b_k.$$

In other words, we generate new entries in the a and b columns is **exactly the same way** we generate new entries in the m column.

Why do we bother with this?

We call numbers m and n *coprime* or *relatively prime* if $\gcd(m, n) = 1$. The extended Euclidean algorithm will then provide a and b with $am + bn = 1$. Therefore $bn \equiv 1 \pmod{m}$. So the extended Euclidean algorithm provides a solution to the congruence $nx \equiv 1 \pmod{m}$.

■ **Example 3.29** Let's solve $71x \equiv 1 \pmod{101}$. The extended Eu-

clidean algorithm gives

$$\gcd(101, 71) = 1 = -26 \times 101 + 37 \times 71.$$

So $37 \times 71 \equiv 1 \pmod{101}$, and $x \equiv 37 \pmod{101}$ is the solution to $71x \equiv 1 \pmod{101}$.

This means that $[37]$ is the reciprocal of $[71]$ in $\mathbb{Z}_{101}$. We can use this to solve other congruences.

■ **Example 3.30** Let us solve $71x \equiv 19 \pmod{101}$. We know $37 \times 71 \equiv 1 \pmod{101}$ from Example 3.29, so

$$x \equiv 37 \times 71x \equiv 37 \times 19 = 703 \equiv 97 \pmod{101}.$$

The solution to the congruence is $x \equiv 97 \pmod{101}$.

It is good to check this:

$$71 \times 97 = 6887 \equiv 19 \pmod{101},$$

just as we need.

We can also derive theoretical consequences. For a start we can completely characterize invertible elements in $\mathbb{Z}_n$.

Theorem 3.31 An element $[a]$ of $\mathbb{Z}_n$ is invertible if and only if $\gcd(a, n) = 1$.

Proof. We already showed in Theorem 3.22 that if $\gcd(a, n) > 1$ then $[a]$ was not invertible.

But we now know that we can solve $ax \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$, that is we can find $[x]$ with $[a][x] = [1]$ in $\mathbb{Z}_n$: $[a]$ is invertible in $\mathbb{Z}_n$. ■

An important special case is when n is a prime number.

Corollary 3.32 If p is a prime number, then every element of $\mathbb{Z}_p$ apart from $[0]$ has an inverse in $\mathbb{Z}_p$

Proof. Let $a \in \mathbb{Z}$. Then $g = \gcd(a, p)$ is a divisor of p , so $g = 1$ or $g = p$. If $g = 1$, then $[a]$ is invertible in $\mathbb{Z}_p$. If $g = p$, then $p \mid a$, so then $[a] = [0]$ in $\mathbb{Z}_p$ (and has no inverse). ■

3.5 Uniqueness of Prime Factorisation

We now return to the uniqueness of prime factorisations, which we stated (but did not prove) in Theorem 3.14.

We need another corollary of Theorem 3.31.

Lemma 3.33 — Euclid's Lemma. If p is a prime, and $p \mid ab$ for some integers a and b , then either $p \mid a$ or $p \mid b$.

Proof. If $p \mid a$, we win, so suppose $p \nmid a$ but $p \mid ab$. Then $[a]$ has an inverse in $\mathbb{Z}_p$, so $[ca] = [c][a] = [1]$ in $\mathbb{Z}_p$ for some c . But $[ab] = [0]$ in $\mathbb{Z}_p$. Therefore

$$[b] = [1][b] = [ca][b] = [cab] = [c][ab] = [c][0] = [0]$$

in $\mathbb{Z}_p$. This means that $p \mid b$. ■

If we prefer, we can write this proof in terms of congruences; we have $ca \equiv 1 \pmod{p}$ and $ab \equiv 0 \pmod{p}$ so that

$$b = 1b \equiv cab \equiv c0 = 0 \pmod{p}.$$

Euclid's lemma extends to products of several numbers. If a prime divides a product of several numbers, it divides one of the numbers. More formally:

Lemma 3.34 Let p be prime, and $a_1, a_2, \dots, a_n \in \mathbb{Z}$. If $p \mid (a_1 a_2 \cdots a_n)$ then $p \mid a_j$ for some j .

Proof. We could phrase this as an induction proof, but I shall be a bit less formal.

As $p \mid (a_1 a_2 \cdots a_n)$, then by Euclid's Lemma, either $p \mid a_1$ or $p \mid (a_2 a_3 \cdots a_n)$. In the former case we win, in the latter case we repeat the argument, so that either $p \mid a_2$ or $p \mid (a_3 \cdots a_n)$. Eventually we find that $p \mid a_j$ for some j . ■

This version of Euclid's Lemma is precisely the tool needed to prove unique factorization into primes. We now repeat, and prove, Theorem 3.14.

Theorem 3.35 — Fundamental Theorem of Arithmetic. Let n have the prime factorizations

$$n = p_1 \cdots p_r = q_1 \cdots q_s.$$

Then every prime occurs equally often in both factorizations.

Proof. Suppose some prime p occurs k times on the left factorization and l times in the right factorization, and that $k \neq l$. I may as well suppose that $k < l$ (otherwise swap the two sides). I can remove k occurrences of p from each side to leave two factorizations of n/p^k :

$$\frac{n}{p^k} = p'_1 \cdots p'_{r-k} = q'_1 \cdots q'_{s-k}.$$

But p remains a prime factor on the right, so that $p \mid q'_1 \cdots q'_{s-k}$ and hence $p \mid p'_1 \cdots p'_{r-k}$. But this contradicts Euclid's lemma as $p \nmid p'_i$ for all i , so p cannot divide the left side $p'_1 \cdots p'_{r-k}$. ■

We now look at an application of unique factorization.

If n is an integer, and p is a prime, define $v_p(n)$ to be the number of times p occurs in the prime factorization of n . For instance if

$n = 13200$, then

$$n = 2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 11 = 2^4 \times 3^1 \times 5^2 \times 11^1.$$

So $v_2(n) = 4$, $v_3(n) = 1$, $v_5(n) = 2$, $v_7(n) = 0$, $v_{11}(n) = 1$, $v_{13}(n) = 0$ and indeed $v_p(n) = 0$ for all other primes. (The v stands for *valuation* but I won't be using this word.)

If we take $m, n \in \mathbb{N}$ and multiply together their prime factorizations, we get a prime factorization of mn with $v_p(m) + v_p(n)$ occurrences of the prime p . As factorization of primes is unique, this number must equal $v_p(mn)$, and we have proved that

$$v_p(mn) = v_p(m) + v_p(n)$$

whenever p is prime.

As an application, we prove that $\sqrt{2}$ is an irrational number.

Were it not, then

$$\sqrt{2} = \frac{m}{n}$$

where $m, n \in \mathbb{N}$. Squaring and re-arranging, gives

$$m^2 = 2n^2.$$

Considering the prime factorization of this number, we see that that of m^2 must contain an even number of twos, while that of $2n^2$ contains an odd number of twos, and this contradicts unique factorization. More formally,

$$v_2(m^2) = v_2(mm) = v_2(m) + v_2(m) = 2v_2(m) \equiv 0 \pmod{2}$$

while

$$v_2(2n^2) = v_2(2nn) = v_2(2) + v_2(n) + v_2(n) = 2v_2(n) + 1 \equiv 1 \pmod{2}.$$

So we cannot have $v_2(m^2) = v_2(2n^2)$. This proves that $\sqrt{2}$ must be irrational.

[The same method can be used to show the irrationality of many other real numbers, e.g. $\sqrt{3}$, $\sqrt[3]{2}$.]

4 GROUPS

4.1 Binary operations

We now take a step back, and look at our subject more abstractly. The natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ all have notions of “binary operations” such as addition, subtraction and multiplication defined on them. A binary operation is an operation that takes two elements and combines them to make a third. More formally:

Definition 4.1 A binary operation $*$ (or just an operation) on a set X is a rule assigning to each $a \in X$ and $b \in X$ an element $a * b \in X$.

As an example, addition $+$ is a binary operation on $\mathbb{N}$. And on $\mathbb{Z}$. And on $\mathbb{Q}$ and $\mathbb{R}$. Why? if a and b are natural numbers, so is $a + b$; if a and b are integers, then so is $a + b$ etc.

After addition, why not subtraction $-$? Alas subtraction is **not** an operation on $\mathbb{N}$: since $3 \in \mathbb{N}$, $7 \in \mathbb{N}$ but $3 - 7 \notin \mathbb{N}$. But subtraction is a binary operation on each of $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.

Multiplication? an operation on each of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.

We also have addition, subtraction and multiplication as binary operations on each $\mathbb{Z}_n$. These are **finite** sets which have interesting binary operations.

Some operations are nicer than others.

Definition 4.2 A binary operation $*$ on a set X is *commutative* if $a * b = b * a$ for all $a, b \in X$.

In all our number systems seen so far, addition and multiplication are commutative: $a + b = b + a$, and $ab = ba$. But subtraction isn't commutative on $\mathbb{Z}$ say, for

$$3 - 5 \neq 5 - 3.$$

Commutativity means not needing to be worried about order in the operation.

Definition 4.3 A binary operation $*$ on a set X is *associative* if $a * (b * c) = (a * b) * c$ for all $a, b, c \in X$.

Associativity is good as it allows us to “drop brackets”; as $a * (b * c)$ and $(a * b) * c$ are the same thing, it’s harmless to write either as $a * b * c$. This extends to four or more elements:

$$a * (b * (c * d)) = a * ((b * c) * d) = (a * (b * c)) * d = \dots$$

In all our number systems so far, addition and multiplication are associative: we can write $a + b + c$ or abc without fear of ambiguity. But subtraction is not associative on $\mathbb{Z}$:

$$(9 - 4) - 2 = 5 - 2 = 3, \text{ but } 9 - (4 - 2) = 9 - 2 = 7.$$

A classic example of an associative operation is *function composition*. Let A be a set and take X to be the set of all mappings from A to A . For $f, g \in X$ we define $g \circ f$ by $(g \circ f)(a) = g(f(a))$. Then, for $f, g, h \in X$,

$$(h \circ (g \circ f))(a) = h((g \circ f)(a)) = h(g(f(a)))$$

and

$$((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a))).$$

Therefore $h \circ (g \circ f)$ and $(h \circ g) \circ f$ define the same function, so they are equal. (We saw this earlier, in a more general setting, in Proposition 2.20.)

4.2 Axioms of a group

These are the axioms for a *group*.

Definition 4.4 A set G with a binary operation $*$ is a group if

1. $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$;
2. there is $e \in G$ with the property that $e * a = a$ and $a * e = a$ for all $a \in G$;
3. for each $a \in G$ there is $a' \in G$ such that $a * a' = e$ and $a' * a = e$.

The operation in a group must be **associative**. Moreover it must possess an **identity** element (an element e as in 2.) and that each element has an **inverse** (in 3., a' is an inverse for a).

We do **not** assume that the group operation is commutative. A group where the operation is commutative is called a *commutative* or an *Abelian* group (after Niels Henrik Abel, 1802–1829).

Of the associative operations we have seen so far, addition is a group operation on $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. We have seen it is associative, and in each of these sets, 0 is the identity, (always $0 + a = a = a + 0$) and the inverse of a is $-a$, (always $a + (-a) = 0 = (-a) + a$). But $\mathbb{N}$ is **not** a group with addition since there is no identity element. Similarly, addition is a group operation on all of the $\mathbb{Z}_n$. Indeed each $\mathbb{Z}_n$ is a **finite** group under addition (it has finitely many elements).

But none of our number systems is a group under multiplication: multiplication is associative, and 1 is the identity element, but 0 has no inverse (in $\mathbb{Z}$, $\mathbb{Q}$, etc.) as $0a = 0 \neq 1$ for all a .

We can get around this by eliminating the zero element. As an example, define

$$\mathbb{Q}^\times = \{a \in \mathbb{Q} : a \neq 0\}.$$

Then multiplication is a binary operation on $\mathbb{Q}^\times$, since if $a, b \in \mathbb{Q}^\times$, ab is rational and nonzero, so $ab \in \mathbb{Q}^\times$. Also 1 is an identity element. Finally, if $a \in \mathbb{Q}^\times$, then $a^{-1} = 1/a$ is a nonzero rational too, and is the inverse of a .

For similar reasons, $\mathbb{R}^\times$, the set of nonzero real numbers is a group with multiplication as the operation.

We could try this trick on $\mathbb{Z}_n$ but we run into difficulties. For $n = 5$, the nonzero elements have this multiplication table:

$\times$	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]
[2]	[2]	[4]	[1]	[3]
[3]	[3]	[1]	[4]	[2]
[4]	[4]	[3]	[2]	[1]

This is fine: we know multiplication is associative, and [1] is the identity. Each row contains a [1], and as multiplication is commutative, then for each [a] there is [b] with $[a][b] = [1] = [b][a]$. We have a group, with four elements, that we'll call $\mathbb{Z}_5^\times$.

But this trick fails in $\mathbb{Z}_6$. Removing the zero element gives a multiplication table

$\times$	[1]	[2]	[3]	[4]	[5]
[1]	[1]	[2]	[3]	[4]	[5]
[2]	[2]	[4]	[0]	[2]	[4]
[3]	[3]	[0]	[3]	[0]	[3]
[4]	[4]	[2]	[0]	[4]	[2]
[5]	[5]	[4]	[3]	[2]	[1]

Associativity and identity are fine, but not all elements have inverses: [2] and [3] and [4] have no inverse. This is not a group. The solution here is to be more radical and throw out all elements without inverses. This leaves the set $\{[1], [5]\}$ with multiplication table

$\times$	[1]	[5]
[1]	[1]	[5]
[5]	[5]	[1]

This is now a group, albeit a puny one.

In general, take $\mathbb{Z}_n$ and consider the elements which have inverses, namely (by Theorem 3.31) the $[a]$ with $\gcd(a, n) = 1$. Call this set $\mathbb{Z}_n^\times$. We claim it is a group. Note that multiplication is a binary operation: if $[a], [b] \in \mathbb{Z}_n^\times$ then $[a], [b]$ have inverses $[a'], [b']$ and then $[a'][b'] = [a'b']$ is an inverse of $[a][b] = [ab]$. Associativity and identity are clear, and as if $[a']$ is the inverse of $[a]$ then $[a]$ is the inverse of $[a']$ then each element of $\mathbb{Z}_n^\times$ has an inverse there.

As an example $\mathbb{Z}_{10}^\times = \{[1], [3], [7], [9]\}$ and has multiplication table

$\times$	[1]	[3]	[7]	[9]
[1]	[1]	[3]	[7]	[9]
[3]	[3]	[9]	[1]	[7]
[7]	[7]	[1]	[9]	[3]
[9]	[9]	[7]	[3]	[1]

All the examples so far have been abelian groups; they obey the commutative law $a * b = b * a$. We have already seen a whole family of non-abelian groups:

■ **Example 4.5** For each $n \in \mathbb{N}$, the set S_n of permutations on $\{1, 2, \dots, n\}$ is a group with the operation $\circ$ (composition of permutations). Indeed, for $\pi, \sigma, \tau \in S_n$ we have $\pi \circ (\sigma \circ \tau) = (\pi \circ \sigma) \circ \tau$ because composition of functions is always associative; the identity permutation satisfies $\text{id} \circ \sigma = \sigma$ and $\sigma \circ \text{id} = \sigma$ by the properties of identity functions; and each $\sigma \in S_n$ has an inverse permutation σ^{-1} with $\sigma \circ \sigma^{-1} = \text{id}$ and $\sigma^{-1} \circ \sigma = \text{id}$. The group S_n has $n!$ elements, and for $n \geq 3$ it is not abelian since, for example if $\sigma = (123)$ and $\tau = (12)$ then

$$\sigma \circ \tau = (123)(12) = (13), \quad \tau \circ \sigma = (12)(123) = (13)$$

so we do not always have $\sigma \circ \tau = \tau \circ \sigma$.

We next note some properties which hold in any group, but are

not actually part of the definition. In order to prove them, all that we can use is what is contained in the definition of a group.

Proposition 4.6 Let G be a group with operation $*$. Then

- (i) G has only one identity element;
- (ii) each $a \in G$ has only one inverse;
- (iii) (left cancellation) if $a, b, x \in G$ and $a * x = a * y$ then $x = y$;
- (iv) (right cancellation) if $a, b, x \in G$ and $x * a = y * a$ then $x = y$.

Proof. (i) Suppose e and f both satisfy the conditions to be an identity element. We must show $e = f$.

For each $a \in G$, we have

$$e * a = a, \quad a * e = a, \quad f * a = a, \quad a * f = a.$$

From the first of these equations (with $a = f$) we have $e * f = f$,

and from the fourth equation (with $a = e$) we have $e * f = e$.

Hence $e = e * f = f$ as required.

- (ii) Suppose a has two inverses a' and $\hat{a}$. We must show $a' = \hat{a}$.

We calculate $a' * (a * \hat{a})$ in two ways. First, since $a * \hat{a} = e$ (because $\hat{a}$ is an inverse to a), we have

$$a' * (a * \hat{a}) = a' * e = a'.$$

Second, using associativity and the fact that $a' * a = e$ (because a' is an inverse to a), we have

$$a' * (a * \hat{a}) = (a' * a) * \hat{a} = e * \hat{a} = \hat{a}.$$

Hence $a' = \hat{a}$.

(iii) Multiplying the given equation $a * x = a * y$ by the inverse a' of a , we have

$$a' * (a * x) = a' * (a * y).$$

Thus, using associativity,

$$(a' * a) * x = (a' * a) * y.$$

But $a' * a = e$, so

$$e * x = e * y$$

and therefore $x = y$.

(iv) Similarly (exercise).



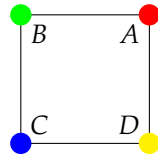
Before proceeding, we note that for most examples of groups, the operation is written as addition or multiplication. We have two main “systems of notation”.

	additive notation	multiplicative notation
$a * b$	$a + b$	ab or $a \times b$
identity	0	e or 1 or id or I
inverse of a	$-a$	a^{-1}
$a * a, a * a * a$ etc.	$2a, 3a$ etc.	a^2, a^3 etc.

We normally only use additive notation for Abelian groups. When dealing with a “generic” group (for instance when we are proving results about all groups) we usually use multiplicative notation.

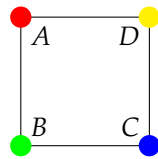
4.3 Dihedral groups

More interesting groups come from geometry. Consider a square sitting in the plane. I’ll label its vertices A, B, C, D as shown.

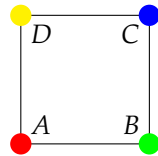


We consider all moves where we take the square, and put it back upon itself. We could rotate it about its centre, or we could reflect the square about one of its axes of symmetry.

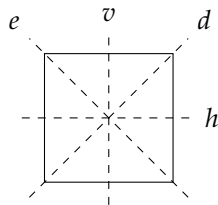
We could rotate the square $90^\circ = \pi/2$ anticlockwise about its centre to give



Or through π anticlockwise (or clockwise) to give

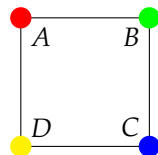


The square has four axes of symmetry, vertical, horizontal, and two diagonal:

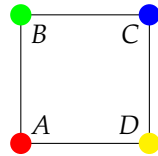


We can reflect (or just pick up the square and turn it over) the square about any of these four axes.

For example reflecting in axis v gives



and reflecting in axis e gives



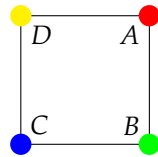
We have a total of 8 *symmetry operations*, four rotations and four reflections.

I'll write the rotations as R_0, R_1, R_2 and R_3 where R_k is the anti-clockwise rotation through $k\pi/2$. I'll write the reflections as S_h, S_v, S_d and S_e where S_h is reflection about the axis h etc.

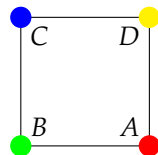
Note that R_0 is the *identity* operation; it leaves the square unchanged.

We can combine symmetry operations: if U and V are symmetry operations then VU is the result first applying operation U , and then operation V .

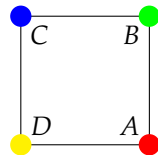
Let's work out R_3S_d . We first apply S_d which takes the square to



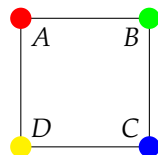
and applying R_3 to this gives



This is the same as a reflection in axis h : $R_3S_d = S_h$. What about S_dR_3 ? Well, R_3 takes the original square to



and then S_d takes this to



The overall effect is reflection in axis v : $S_d R_3 = S_v$. So $S_d R_3 \neq R_3 S_d$: the operation is not commutative.

Let $G = \{R_0, R_1, R_2, R_3, S_h, S_d, S_v, S_e\}$. Is G a group? One finds that each $VU \in G$ for each U and $V \in G$. Is multiplication associative? Any way $W(VU)$ and $(WV)U$ both are the results of applying operation U , then V and finally W , so yes it is associative.

The identity is R_0 . Finally most elements are their own inverses. The exceptions are R_1 and R_3 which are inverses of each other.

Thus G is a non-Abelian group of eight elements.

Here is the complete multiplication table for G

	R_0	R_1	R_2	R_3	S_h	S_d	S_v	S_e
R_0	R_0	R_1	R_2	R_3	S_h	S_d	S_v	S_e
R_1	R_1	R_2	R_3	R_0	S_d	S_v	S_e	S_h
R_2	R_2	R_3	R_0	R_1	S_v	S_e	S_h	S_d
R_3	R_3	R_0	R_1	R_2	S_e	S_h	S_d	S_v
S_h	S_h	S_e	S_v	S_d	R_0	R_3	R_2	R_1
S_d	S_d	S_h	S_e	S_v	R_1	R_0	R_3	R_2
S_v	S_v	S_d	S_h	S_e	R_2	R_1	R_0	R_3
S_e	S_e	S_v	S_d	S_h	R_3	R_2	R_1	R_0

Some observations: the table is asymmetric about main diagonal (group is non-Abelian). The product of two R s or two S s is an R , and the product of an R and an S (either way) is an S .

This symmetry group of the square is the *dihedral group of order 8*. Likewise the symmetry group of a regular polygon with n edges has n rotations and n reflections and is called the dihedral group of order $2n$. Some authors name this group D_n , and some name it D_{2n} . I will call it D_{2n} . (So the symmetry group of the square is D_8 in my notation.)

We can express all elements of our group G in terms of the elements R_1 and S_h . Let's write R for R_1 and S for S_h . Also write I

for R_0 , the identity element. Then $R_2 = R^2$, $R_3 = R^3$, $R_0 = I = R^4$, $S_d = RS$, $S_v = R^2S$, $S_e = R^3S$. So every element of G is an R^k or an R^kS . Then $S^2 = I$ but $SR = S_hR_1 = S_e = R^3S = R^{-1}S$. It follows that $SR^2 = R^{-1}SR = R^{-2}S$, and by an easy induction $SR^k = R^{-k}S$.

Using the equations

$$R^4 = I, \quad S^2 = I \quad SR = R^{-1}S$$

we can compute the product of any two elements of the group.

For example,

$$(R^3S)(R^2S) = R^3SR^2S = R^3R^{-1}SRS = R^2R^{-1}S^2 = RI = R.$$

In this notation the group multiplication table of D_8 becomes:

	I	R	R^2	R^3	S	RS	R^2S	R^3S
I	I	R	R^2	R^3	S	RS	R^2S	R^3S
R	R	R^2	R^3	I	RS	R^2S	R^3S	S
R^2	R^2	R^3	I	R	R^2S	R^3S	S	RS
R^3	R^3	I	R	R^2	R^3S	S	RS	R^2S
S	S	R^3S	R^2S	RS	I	R^3	R^2	R
RS	RS	S	R^3S	R^2S	R	I	R^3	R^2
R^2S	R^2S	RS	S	R^3S	R^2	R	I	R^3
R^3S	R^3S	R^2S	RS	S	R^3	R^2	R	I

4.4 The cardinality of a group and the order of an element

So far we have concentrated on examples of groups. It is time to develop some theory. From now on we write the typical group multiplicatively, with identity e .

Definition 4.7 The *cardinality* (or *order*) of a group is the number of elements it has. If this number is finite, it is a *finite group* and otherwise an *infinite group*. We write $|G|$ for the cardinality of G .

As examples, the additive groups $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$, and the multiplicative groups $\mathbb{Q}^\times$ and $\mathbb{R}^\times$ are infinite groups.

Always $|\mathbb{Z}_n| = n$ but the order of $\mathbb{Z}_n^\times$ varies; when n is prime $|\mathbb{Z}_n^\times| = n - 1$ but $|\mathbb{Z}_6^\times| = 2$ and $|\mathbb{Z}_{10}^\times| = 4$ etc.

The cardinality of the symmetry group of the square is 8, and that of the regular n -gon is $2n$ (we call this group D_{2n}).

■ **Example 4.8** We saw in §2.4 that the cardinality of S_n is $n!$.

We can combine a group element with itself, once or twice or many times. If G is a group, written multiplicatively and $g \in G$ then we can consider gg which we write as g^2 for obvious reasons. We can take this further: $g^3 = g(gg) = (gg)g$ etc. In an additive group we consider $g + g$, $g + g + g$ etc., which we write as $2g$, $3g$ etc. We can therefore define g^n for $n \in \mathbb{N}$ (or ng in an additive group). Naturally we define $g^0 = e$ and g^{-n} to be the inverse of g^n (or $0g = 0$ and $(-n)g$ as the inverse of ng in an additive group.)

Lemma 4.9 We have $g^m g^n = g^{m+n}$ and $(g^m)^n = g^{mn}$ for all $m, n \in \mathbb{Z}$.

I won't prove this, as the proof is fiddly and involves splitting into a lot of separate cases, but for small values of m and n it is easy to verify.

As examples, in the additive group $\mathbb{Z}_7$ we have $2[2] = [4]$, $3[2] = [6]$, $4[2] = [1]$ etc., and in the multiplicative group $\mathbb{Z}_7^\times$ we have $[2]^2 = [4]$, $[2]^3 = [1]$, $[2]^4 = [2]$ etc. After this, the series of powers of $[2]$ repeats.

I claim this repetition is typical behaviour in finite groups. If $g \in G$ with G finite, the sequence

$$g, g^2, g^3, \dots,$$

cannot contain an infinite number of elements. There must be a

repeat at some stage: $g^b = g^a$ where $a < b$. Then $g^{b-a} = g^b g^{-a} = g^b (g^a)^{-1} = g^b (g^b)^{-1} = e$. Thus $g^r = e$ for some positive integer r . The least positive integer such that $g^r = e$ is the *order* of g and denoted by $o(g)$.

In an infinite group, there may or may not be a positive integer r with $g^r = e$. For example in $\mathbb{R}^\times$, $(-1)^2 = 1$ is the identity, so -1 has *finite order* in $\mathbb{R}^\times$, in this case $o(-1) = 2$. But $2^r \neq 1$ for all $r \in \mathbb{N}$, so 2 has *infinite order* in $\mathbb{R}^\times$.

In summary:

Definition 4.10 Let G be a group (written multiplicatively) and let $g \in G$. The *order* of g , written $o(g)$ is the least positive integer r with $g^r = e$, if such an r exists. Otherwise, we write $o(g) = \infty$ and say that g has *infinite order*.

We have already seen this idea in the case of permutations in Definition 2.41.

In S_n there is a useful formula for the order of an element, as we saw in Lemma 2.42: the order of σ is the least common multiple of its cycle lengths since if j is a number within a length d cycle of σ , then $\sigma^r(j) = j$ if and only if $d \mid r$.

■ **Example 4.11**

$$(137)(29)(4586)$$

has order 12 in S_9 as 12 is the least common multiple of 3, 2 and 4.

4.5 Subgroups

A *subgroup* is a group within a group. More precisely:

Definition 4.12 If G is a group with operation $*$ then a subset H of G is a *subgroup* of G if

- $a * b \in H$ whenever $a \in H$ and $b \in H$,

- the identity element of G is an element of H ,
- the inverse of each element of H is an element of H .

It's quite straightforward to see that a subgroup H of a group G is a group under the "same" operation. Associativity holds for H since it holds inside G , and identity and inverses are guaranteed by the above conditions.

We write $H \leq G$ as shorthand for " H is a subgroup of G ."

■ **Example 4.13** Consider the group D_8 of symmetry operations of a square with multiplication table

	I	R	R^2	R^3	S	RS	R^2S	R^3S
I	I	R	R^2	R^3	S	RS	R^2S	R^3S
R	R	R^2	R^3	I	RS	R^2S	R^3S	S
R^2	R^2	R^3	I	R	R^2S	R^3S	S	RS
R^3	R^3	I	R	R^2	R^3S	S	RS	R^2S
S	S	R^3S	R^2S	RS	I	R^3	R^2	R
RS	RS	S	R^3S	R^2S	R	I	R^3	R^2
R^2S	R^2S	RS	S	R^3S	R^2	R	I	R^3
R^3S	R^3S	R^2S	RS	S	R^3	R^2	R	I

Inside this group the subset $H = \{I, R, R^2, R^3\}$ of rotations forms a subgroup. The group table for this subgroup is:

	I	R	R^2	R^3	S	RS	R^2S	R^3S
I	I	R	R^2	R^3	S	RS	R^2S	R^3S
R	R	R^2	R^3	I	RS	R^2S	R^3S	S
R^2	R^2	R^3	I	R	R^2S	R^3S	S	RS
R^3	R^3	I	R	R^2	R^3S	S	RS	R^2S
S	S	R^3S	R^2S	RS	I	R^3	R^2	R
RS	RS	S	R^3S	R^2S	R	I	R^3	R^2
R^2S	R^2S	RS	S	R^3S	R^2	R	I	R^3
R^3S	R^3S	R^2S	RS	S	R^3	R^2	R	I

But the set of reflections $K = \{S, RS, R^2S, R^3S\}$ do not; for instance $S(RS) = R^3 \notin K$.

■ **Example 4.14** We know $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is a group under addition. $\mathbb{N} = \{1, 2, 3, \dots\}$ is closed under addition ($a, b \in \mathbb{N}$ implies $a + b \in \mathbb{N}$) but $\mathbb{N}$ lacks inverses. So $\mathbb{N}$ is *not* a subgroup of $\mathbb{Z}$.

But $2\mathbb{Z} = \{2a : a \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$. This is the set of even integers. Adding two even integers gives an even integer; zero is an even integer, and the negative of an even integer is even. So $2\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.

An important class of subgroups are the *alternating groups*. Consider S_n . Recall from Theorem 2.45 that each permutation in S_n is either even or odd (but not both).

■ **Example 4.15** Let

$$A_n = \{\sigma \in S_n : \sigma \text{ is an even permutation}\}.$$

Then A_n is a subgroup of S_n , the *alternating group of degree n* .

To prove that A_n is a subgroup, we first note that the product of two even permutations is even, and that the identity permutation is even. We need to prove that each even permutation has an even inverse, otherwise there is a permutation with σ even but σ^{-1} odd. That means that $\sigma\sigma^{-1}$ is an odd permutation, but $\sigma\sigma^{-1} = \text{id}$ which is an even permutation. So indeed A_n is a subgroup of S_n .

For example,

$$A_3 = \{\text{id}, (123), (132)\}.$$

There are some boring examples of subgroups too. If G is a group with identity e , then G is a subgroup of G , and also $\{e\}$ is a subgroup of G . When $H \leq G$ we say H is a *proper subgroup* of G if $H \neq G$ and

that H is a *non-trivial subgroup* of G if $H \neq \{e\}$.

Apart from these, the simplest possible subgroups of a group G are *cyclic subgroups*.

Definition 4.16 If $a \in G$, then the set $\{a^n : n \in \mathbb{Z}\}$ is a subgroup of G , which we denote by $\langle a \rangle$. This is the *cyclic subgroup* of G generated by a .

As examples, in D_8 , $\langle R \rangle = \{I, R, R^2, R^3\} = \langle R^3 \rangle$, $\langle R^2 \rangle = \{I, R^2\}$, $\langle I \rangle = \{I\}$ and for each k , $\langle R^k S \rangle = \{I, R^k S\}$. So D_8 has seven cyclic subgroups.

Definition 4.17 A group is *cyclic* if it is a cyclic subgroup of itself, that means it has a *generator* a and all its elements are powers of a .

Every group $\mathbb{Z}_n$ is cyclic. This is an additive group, and in additive groups, $\langle a \rangle = \{na : n \in \mathbb{Z}\}$. In all cases $\mathbb{Z}_n = \langle [1]_n \rangle$.

Also $\mathbb{Z} = \langle 1 \rangle$ is a cyclic group. But $\mathbb{Q}$ is not a cyclic group. If, say $\mathbb{Q} = \langle a \rangle$ then $1 = na$ for some $n \in \mathbb{Z}$. Then $n \neq 0$ and $a = 1/n$. But then $1/(2n) \notin \langle 1/n \rangle$.

Remark 4.18

In general, if $a \in G$ is an element with finite order m , then

$$\langle a \rangle = \{e, a, a^2, \dots, a^{m-1}\}$$

has exactly m elements. Therefore $o(a) = |\langle a \rangle|$. This relates the two uses of the word “order” in group theory: the order of an element is the same as the cardinality (order) of the cyclic subgroup it generates.

We have seen D_8 has cyclic subgroups of cardinality 1, 2 and 4. It also has non-cyclic subgroups of cardinality 4, for instance $H = \{I, R^2, S, R^2 S\}$.

Here is the group table for $H = \{I, R^2, S, R^2S\}$.

	I	R	R^2	R^3	S	RS	R^2S	R^3S
I	I	R	R^2	R^3	S	RS	R^2S	R^3S
R	R	R^2	R^3	I	RS	R^2S	R^3S	S
R^2	R^2	R^3	I	R	R^2S	R^3S	S	RS
R^3	R^3	I	R	R^2	R^3S	S	RS	R^2S
S	S	R^3S	R^2S	RS	I	R^3	R^2	R
RS	RS	S	R^3S	R^2S	R	I	R^3	R^2
R^2S	R^2S	RS	S	R^3S	R^2	R	I	R^3
R^3S	R^3S	R^2S	RS	S	R^3	R^2	R	I

We observe that the cardinality of each subgroup is a factor of the cardinality of D_8 , that is 8. Is this always true? Is the cardinality of a subgroup of a finite group a factor of the cardinality of the group?

4.6 Cosets and Lagrange's Theorem

Our goal in this section is to prove the following important result:

Theorem 4.19 — Lagrange's theorem. Let G be a finite group, and let H be a subgroup of G . Then $|H|$ is a factor of $|G|$.

(Joseph-Louis Lagrange, 1736–1813)

To prove this, we need the idea of *cosets*. Throughout this section, G is a finite group (written multiplicatively) and H is a subgroup of G .

Definition 4.20 For $g \in G$, we write

$$gH = \{gh : h \in H\}.$$

So gH is a *subset* (not subgroup) of G , and it contains precisely $|H|$ elements, since if $gh = gh'$ then (multiplying on the left by g^{-1}) we get $h = h'$.

We call gH the **left coset** of H determined by g .

■ **Example 4.21** Let $G = D_8 = \{I, R, R^2, R^3, S, RS, R^2S, R^3S\}$ and let H be the subgroup $\{I, S\}$. We have a coset gH for each of the 8 elements $g \in D_8$:

$$IH = \{II, IS\} = \{I, S\} = H.$$

$$RH = \{RI, RS\} = \{R, RS\}.$$

$$R^2H = \{R^2I, R^2S\} = \{R^2, R^2S\}.$$

$$R^3H = \{R^3I, R^3S\} = \{R^3, R^3S\}.$$

$$SH = \{SI, SS\} = \{S, I\} = H.$$

$$(RS)H = \{(RS)I, (RS)S\} = \{RS, R\}.$$

$$(R^2S)H = \{(R^2S)I, (R^2S)S\} = \{R^2S, R^2\}.$$

$$(R^3S)H = \{(R^3S)I, (R^3S)S\} = \{R^3S, R^3\}.$$

We only get 4 different cosets:

$$\{I, S\} = H, \quad \{R, RS\}, \quad \{R^2, R^2S\}, \quad \{R^3, R^3S\},$$

each of which occurs as gH for exactly two elements g . Moreover, each element of D_8 occurs in exactly one of these 4 cosets.

Remark 4.22

Instead of left cosets gH , we could consider right cosets,

$$Hg = \{hg : h \in H\}.$$

But right cosets and left cosets need not agree, e.g. $HR = \{R, SR\} = \{R, R^3S\}$, since $SR = R^3S$ in D_8 , but this set does not appear in our list of 4 left cosets.

Proof of Lagrange's Theorem. Recall that H is a subgroup in the finite group G .

Let $g_1H, g_2H, \dots, g_rH$ be the *different* left cosets gH .

Each of these cosets has size $|H|$, and every element $g \in G$ is in at least one of the cosets: gH must appear in our list, so $gH = g_iH$ for some i , and $g = ge \in gH$. The crucial point is that each g occurs in *only* one coset g_iH .

For if $g \in g_iH \cap g_jH$ then we can write $g = g_ih$ and $g = g_jh'$ for $h, h' \in H$. Then $g_ih = g_jh'$ and so $g_i = g_jh'h^{-1}$. If $x \in g_iH$, we can write $x = g_ih^*$ for some $h^* \in H$, and then $x = (g_jh'h^{-1})h^* = g_j(h'h^{-1}h^*) \in g_jH$. This shows that $g_iH \subseteq g_jH$. But these sets both have size $|H|$, so $g_iH = g_jH$. So each $g \in G$ is contained in exactly one of the cosets $g_1H, \dots, g_rH$.

It follows that

$$|G| = |g_1H| + \dots + |g_rH| = r|H|,$$

so that $|H|$ is a factor of $|G|$. ■

■ **Example 4.23** If we apply Lagrange's theorem to a cyclic subgroup $\langle a \rangle$ of a finite group G , we find that $o(a)$ is a factor of $|G|$. This is because $o(a) = |\langle a \rangle|$. Moreover, if we write $k = o(a)$, $n = |G|$ and

$r = n/k$, which is an integer, then

$$a^{|G|} = a^{kr} = (a^k)^r = e^r = e.$$

This is often a useful fact. As an example, if $\sigma \in S_n$ then $\sigma^{n!} = \text{id}$.

■ **Example 4.24** As another example, if p is prime then $\mathbb{Z}_p^\times$ is a group of cardinality $p - 1$. If $a \in \mathbb{Z}$ and $p \nmid a$ then $[a] \in \mathbb{Z}_p^\times$. Therefore $[a]^{p-1} = [1]$ in $\mathbb{Z}_p^\times$, that is

$$a^{p-1} \equiv 1 \pmod{p}.$$

This is *Fermat's Little Theorem*.

As the contrapositive of this, note that if $n \nmid a$ and $a^{n-1} \not\equiv 1 \pmod{n}$ then n cannot be a prime number. This observation is the basis of modern tests for primality of numbers.

4.7 Isomorphisms and homomorphisms

We now look at *isomorphism* of groups. The idea is to determine when two groups have the “same” structure, or that they are the same group with elements labelled differently.

■ **Example 4.25** Consider $G = \mathbb{Z}_7^\times$. It has six elements, with multiplication table:

$\times$	$[1]_7$	$[2]_7$	$[3]_7$	$[4]_7$	$[5]_7$	$[6]_7$
$[1]_7$	$[1]_7$	$[2]_7$	$[3]_7$	$[4]_7$	$[5]_7$	$[6]_7$
$[2]_7$	$[2]_7$	$[4]_7$	$[6]_7$	$[1]_7$	$[3]_7$	$[5]_7$
$[3]_7$	$[3]_7$	$[6]_7$	$[2]_7$	$[5]_7$	$[1]_7$	$[4]_7$
$[4]_7$	$[4]_7$	$[1]_7$	$[5]_7$	$[2]_7$	$[6]_7$	$[3]_7$
$[5]_7$	$[5]_7$	$[3]_7$	$[1]_7$	$[6]_7$	$[4]_7$	$[2]_7$
$[6]_7$	$[6]_7$	$[5]_7$	$[4]_7$	$[3]_7$	$[2]_7$	$[1]_7$

This table may look a jumble, but G is a cyclic group with $[3]_7$ as a generator.

We have $[3]_7^0 = [1]_7$, $[3]_7^1 = [3]_7$, $[3]_7^2 = [2]_7$, $[3]_7^3 = [6]_7$, $[3]_7^4 = [4]_7$,
 $[3]_7^5 = [5]_7$ (and $[3]_7^6 = [1]_7$). We can write the table as follows:

$\times$	$[3]_7^0$	$[3]_7^2$	$[3]_7^1$	$[3]_7^4$	$[3]_7^5$	$[3]_7^3$
$[3]_7^0$	$[3]_7^0$	$[3]_7^2$	$[3]_7^1$	$[3]_7^4$	$[3]_7^5$	$[3]_7^3$
$[3]_7^2$	$[3]_7^2$	$[3]_7^4$	$[3]_7^3$	$[3]_7^0$	$[3]_7^1$	$[3]_7^5$
$[3]_7^1$	$[3]_7^1$	$[3]_7^3$	$[3]_7^2$	$[3]_7^5$	$[3]_7^0$	$[3]_7^4$
$[3]_7^4$	$[3]_7^4$	$[3]_7^0$	$[3]_7^5$	$[3]_7^2$	$[3]_7^3$	$[3]_7^1$
$[3]_7^5$	$[3]_7^5$	$[3]_7^1$	$[3]_7^0$	$[3]_7^3$	$[3]_7^4$	$[3]_7^2$
$[3]_7^3$	$[3]_7^3$	$[3]_7^5$	$[3]_7^4$	$[3]_7^1$	$[3]_7^2$	$[3]_7^0$

This is looking very cluttered: let's write t for $[3]_7$ (and e for the identity).

$\times$	e	t^2	t	t^4	t^5	t^3
e	e	t^2	t	t^4	t^5	t^3
t^2	t^2	t^4	t^3	e	t	t^5
t	t	t^3	t^2	t^5	e	t^4
t^4	t^4	e	t^5	t^2	t^3	t
t^5	t^5	t	e	t^3	t^4	t^2
t^3	t^3	t^5	t^4	t	t^2	e

Permuting rows and columns gives

$\times$	e	t	t^2	t^3	t^4	t^5
e	e	t	t^2	t^3	t^4	t^5
t	t	t^2	t^3	t^4	t^5	e
t^2	t^2	t^3	t^4	t^5	e	t
t^3	t^3	t^4	t^5	e	t	t^2
t^4	t^4	t^5	e	t	t^2	t^3
t^5	t^5	e	t	t^2	t^3	t^4

which is very similar to the addition table of $\mathbb{Z}_6$:

+	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$
$[0]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$
$[1]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$
$[2]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$
$[3]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$
$[4]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$
$[5]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$

This is the addition table for the group $\mathbb{Z}_6$. Making the substitution $\phi : [k]_6 \mapsto t^k = [3]_7^k$ takes the addition table of $\mathbb{Z}_6$ to the multiplication table of the group $\mathbb{Z}_7^\times$. In effect, $\mathbb{Z}_6$ and $\mathbb{Z}_7^\times$ are the “same” group; it’s just the elements have been relabelled, and the notation for the operation has changed (from addition to multiplication).

This map ϕ is an example of an *isomorphism* of groups.

Definition 4.26 Let G and H be groups with operations denoted by $*$ and $\bullet$ respectively. An **isomorphism** from G to H is a bijection $\phi : G \rightarrow H$ which is “structure-preserving” in the sense that

$$\phi(a * b) = \phi(a) \bullet \phi(b)$$

for all $a, b \in G$.

Groups G and H are *isomorphic* (and we write $G \cong H$) if there is an isomorphism $\phi : G \rightarrow H$.

So, if the operation in G is addition and the operation in H is multiplication we require

$$\phi(a + b) = \phi(a)\phi(b).$$

If the operation in G is multiplication and the operation in H is

addition we require

$$\phi(ab) = \phi(a) + \phi(b)$$

etc.

In our example we have $\phi : \mathbb{Z}_6 \rightarrow \mathbb{Z}_7^\times$ defined as follows.

a	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$
$\phi(a)$	$[1]_7$	$[3]_7$	$[2]_7$	$[6]_7$	$[4]_7$	$[5]_7$

In short, $\phi([a]_6) = t^a$ for $a \in \{0, 1, 2, 3, 4, 5\}$, where $t = [3]_7$.

Clearly, since ϕ is bijective, all we need to prove that ϕ is an isomorphism is that $\phi([a] + [b]) = \phi([a])\phi([b])$ for all $[a]$ and $[b] \in \mathbb{Z}_6$.

We can take $a, b \in \{0, 1, 2, 3, 4, 5\}$ and then $[a] + [b] = [c]$ where $c \in \{0, 1, 2, 3, 4, 5\}$ and also either $c = a + b$ or $c = a + b - 6$. If $c = a + b$, then

$$\phi([a] + [b]) = \phi([c]) = t^c = t^{a+b} = t^a t^b = \phi([a])\phi([b]).$$

If $c = a + b - 6$, then

$$\phi([a] + [b]) = \phi([c]) = t^c = t^{a+b-6} = t^a t^b (t^6)^{-1} = \phi([a])\phi([b])$$

since t^6 is the identity.

Therefore ϕ really is an isomorphism. Essentially the same argument proves that if $H = \langle h \rangle$ is a cyclic group of cardinality n , then $\phi : [a]_n \mapsto h^a$ is an isomorphism between $\mathbb{Z}_n$ and H .

As isomorphism of groups is intended to reflect that groups have the same “structure”, one expects that isomorphism should be an equivalence relation, that is $G \cong G$ for all G , $G \cong H$ implies $H \cong G$, and $G \cong H$ and $H \cong K$ imply together that $G \cong K$.

It is straightforward to see that the identity map on G is an isomorphism from G to itself. Moreover if $\phi : G \rightarrow H$ and $\psi : H \rightarrow K$

are isomorphisms, then so is $\psi \circ \phi : G \rightarrow K$. So the reflexivity and transitivity of isomorphism is relatively straightforward. However symmetry is the most subtle. To prove this it suffices to prove that if $\phi : G \rightarrow H$ is an isomorphism, then so is $\phi^{-1} : H \rightarrow G$.

Theorem 4.27 Let $\phi : G \rightarrow H$ be an isomorphism of groups. Then $\phi^{-1} : H \rightarrow G$ is also an isomorphism of groups.

Proof. I'll write both groups in multiplicative notation. As ϕ is a bijection, it certainly has an inverse function $\phi^{-1} : H \rightarrow G$ (see Theorem 2.27) which is also a bijection. Let $a, b \in H$. Then $a = \phi(\phi^{-1}(a))$ and $b = \phi(\phi^{-1}(b))$. As ϕ is an isomorphism then

$$ab = \phi(\phi^{-1}(a))\phi(\phi^{-1}(b)) = \phi(\phi^{-1}(a)\phi^{-1}(b)).$$

Applying ϕ^{-1} gives

$$\phi^{-1}(ab) = \phi^{-1}(\phi(\phi^{-1}(a)\phi^{-1}(b))) = \phi^{-1}(a)\phi^{-1}(b).$$

As ϕ^{-1} is bijective, then ϕ^{-1} is an isomorphism. ■

It's convenient to loosen the definition of isomorphism somewhat. If we drop the requirement that the map is bijective we get the following:

Definition 4.28 Let G and H be groups with operations denoted by $*$ and $\bullet$ respectively. A **homomorphism** from G to H is a map $\phi : G \rightarrow H$ which satisfies

$$\phi(a * b) = \phi(a) \bullet \phi(b)$$

for all $a, b \in G$.

There are plenty of examples of homomorphisms. If we take G and H to be any groups, then $\phi(a) = e$ is a rather trivial example of

a homomorphism. This is usually neither injective nor surjective.

We soon start to see homomorphisms everywhere.

- If G is a group and $a \in G$ then $\phi : n \mapsto a^n$ is a homomorphism from $\mathbb{Z}$ to G .
- If $G = \mathbb{R}$ (as an additive group) and $H = \mathbb{R}^\times$ (as a multiplicative group) then $\phi(x) = e^x$ defines a homomorphism from $\mathbb{R}$ to $\mathbb{R}^\times$. This requires $\phi(x+y) = \phi(x)\phi(y)$, which is true as $e^{x+y} = e^x e^y$.
- For $n \in \mathbb{N}$, the map $\phi : a \mapsto [a]_n$ gives a homomorphism from $\mathbb{Z}$ to $\mathbb{Z}_n$.
- $\phi(z) = \log |z|$ defines a homomorphism from $\mathbb{R}^\times$ to $\mathbb{R}$.
- Consider $\text{sgn} : S_n \rightarrow \mathbb{Q}^\times$. This is a homomorphism, as $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$ for all σ and τ .

If $\phi : G \rightarrow H$ is a homomorphism, its *image* is its range as a function namely $\phi(G) = \{\phi(a) : a \in G\}$.

Theorem 4.29 If $\phi : G \rightarrow H$ is a homomorphism, then its image is a subgroup of H .

Proof. I'll write both groups multiplicatively, with e as the identity in both cases.

Let $K = \phi(G)$. Then K is closed under multiplication: the typical elements of K are $\phi(a)$ and $\phi(b)$ with $a, b \in G$, and then $\phi(a)\phi(b) = \phi(ab) \in K$. We need also to show that $e \in K$ and that K is closed under taking inverses. It suffices to prove that $\phi(e) = e$ and $\phi(a)^{-1} = \phi(a^{-1})$.

Now $\phi(e) = \phi(ee) = \phi(e)\phi(e)$ and multiplying by $\phi(e)^{-1}$ gives $e = \phi(e)$. Also $e = \phi(e) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$. Multiplying on the left by $\phi(a)^{-1}$ gives $\phi(a)^{-1} = \phi(a^{-1})$. ■

Within the previous proof, we proved in passing, that homomorphisms preserve identities, and also respect inverses.

In detail if $\phi : G \rightarrow H$ is a homomorphism, then $\phi(e) = e$ and $\phi(a^{-1}) = \phi(a)^{-1}$.

Here we are writing the group operation as multiplication. But if say, G was an additive group and H a multiplicative group (so that $\phi(a + b) = \phi(a)\phi(b)$), this amounts to $\phi(0) = 1$ and $\phi(-a) = \phi(a)^{-1}$.

A homomorphism need not be injective or surjective. If it is both, then it is an isomorphism. But if it is injective, it is almost an isomorphism.

Theorem 4.30 Let $\phi : G \rightarrow H$ be an injective homomorphism. Then $\phi_0 : G \rightarrow \phi(G)$ defined by $\phi_0(a) = \phi(a)$ is an isomorphism between G and the image $\phi(G)$ of G .

Proof. As ϕ_0 is basically the “same” function as ϕ (only the codomain is different) then it is a homomorphism from G to $\phi(G)$. It is still injective, but restricting the codomain makes it surjective too. So ϕ_0 is a bijective homomorphism, that is an isomorphism. ■

We finish this section of the course with a big theorem. Cayley’s theorem states in effect that every finite subgroup is isomorphic to a subgroup of some group S_n . More precisely

Theorem 4.31 — Cayley. Let G be a finite group of cardinality n . Then G is isomorphic to a subgroup of S_n .

(Arthur Cayley, 1821–1895)

This is an example of a *representation theorem*. It represents an abstract structure (a finite group subject to the group axioms) in a concrete guise (as a subgroup of a group S_n , in which we can write down and calculate with its elements).

Proof. It suffices to find an *injective* homomorphism from G to S_n .

Then its image $\phi(G)$ will be a subgroup of S_n and will be isomorphic to G .

The key to the proof is that multiplying by a fixed element of a permutes the elements of G . Label the n elements of G as $g_1, g_2, \dots, g_n$. Let $a \in G$. Then $ag_1, ag_2, \dots, ag_n$ is also a list of the elements of G , each element occurring once each. For a typical element ag_i we have $ag_i = g_j$ for some j depending on i . Write $\rho(i)$ for this j , that is

$$ag_i = g_{\rho(i)}.$$

Then ρ is a mapping from $[n]$ to $[n]$, and as each element $g_j \in G$ is ag_i for a unique i , then ρ is bijective. That is, $\rho \in S_n$. This $\rho \in S_n$ depends on $a \in G$.

Define $\phi : G \rightarrow S_n$ by $\phi(a) = \rho$ where $ag_i = g_{\rho(i)}$.

We claim that ϕ is a group homomorphism, and that it is injective. Let $a, b \in G$ and let $\rho = \phi(a)$, $\sigma = \phi(b)$ and $\tau = \phi(ab)$. To prove ϕ is a homomorphism all we have to do is to prove that $\tau = \rho\sigma$. By definition, for each i ,

$$ag_i = g_{\rho(i)}, \quad bg_i = g_{\sigma(i)}, \quad abg_i = g_{\tau(i)}.$$

But also

$$abg_i = ag_{\sigma(i)} = g_{\rho(\sigma(i))} = g_{\rho\sigma(i)}.$$

Therefore $\tau = \rho\sigma$. This proves that ϕ is a homomorphism.

To show that ϕ is injective, suppose that $\phi(a) = \phi(b) = \rho$. Then $ag_i = g_{\rho(i)} = bg_i$ for all i . For some i , $g_i = e$ is the identity, and so $ae = be$, that is $a = b$. So ϕ is injective, and this completes the proof. ■

■ **Example 4.32** As an example of this construction, take $G = D_8$, the group of symmetries of the square. Write $g_1 = I$, $g_2 = R$, $g_3 = R^2$,

$g_4 = R^3, g_5 = S, g_6 = RS, g_7 = R^2S$ and $g_8 = R^3S$. Then $Rg_1 = RI = R = g_2, Rg_2 = RR = R^2 = g_3$ etc. Let $\rho = \phi(R)$, then $\rho(1) = 2, \rho(2) = 3$ etc. We eventually get

$$\rho = (1234)(5678).$$

We can see this by looking at the R -row of the multiplication table for D_8 :

	I	R	R^2	R^3	S	RS	R^2S	R^3S
R	R	R^2	R^3	I	RS	R^2S	R^3S	S

which we can rewrite as

	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
R	g_2	g_3	g_4	g_1	g_6	g_7	g_8	g_5

In this notation, the S -row is

	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
S	g_5	g_8	g_7	g_6	g_1	g_4	g_3	g_2

This means that $\phi(S) = \sigma = (15)(28)(37)(46)$.

So $\phi(RS) = \phi(R)\phi(S) = \rho\sigma = (16)(25)(38)(47)$ etc. We may check this with the table:

	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
RS	g_6	g_5	g_8	g_7	g_2	g_1	g_4	g_3

Index

- Abelian group, 96
- alternativng group, 107
- associative, 95

- Bezout's formula, 87
- binary operation, 94

- cardinality of a group, 103
- Cayley's theorem, 118
- commutative, 94
- composition of functions, 95
- congruent, 76
- coprime, 88
- cyclic group, 108
- cyclic subgroup, 107

- dihedral group, 99, 102
- divides, 70
- divisible, 70

- Euclid's Lemma, 90
- even permutation, 59
- extended Euclidean algorithm, 88

- factor, 70
- Fermat's Little Theorem, 112
- finite group, 103
- finite order, 104

- generator, 108
- greatest common divisor, 83
- group, 96

- highest common factor, 83
- homomorphism, 116

- identity element, 96
- image, 117
- infinite group, 103
- infinite order, 104
- integers, 61
- integers modulo n , 80
- inverse, 96
- isomorphism, 112, 114

- Lagrange's theorem, 109

- modular arithmetic, 76
- multiple, 70

- natural numbers, 61
- non-trivial subgroup, 107

- odd permutation, 59
- order, 103
- order of an element, 104

- prime number, 69
- proper subgroup, 107

- quotient, 78

- relatively prime, 88
- remainder, 78

- Sieve of Eratosthenes, 71
- sign of a permutation, 60
- subgroup, 105